Class 10
MATHEMATICS

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1. BINOMIAL THEOREM

WORKSHEET-1

Level – I

1. B	2. C	3. A	4. D	5. B	6. B	7. D
8. B	9. B	10. B				

Level – II

1. B	2. D	3. A	4. C	5. D	6.C	7.A
8. 21	9. C	10. D	11. C	12. 1	13.C	14.C
15.A	16. 1	17. A	18. D	19. 7	20.A	21.1

1. [2]

2. [4] Sol: The middle term in the expansion $\left(x-\frac{1}{2x}\right)^{10}$

$= \mathrm{T}_{\frac{10}{2}+1} = \mathrm{T}_6 = {}^{10}\mathrm{C}_5 \mathrm{x}^{10-5}\left(\frac{-1}{2\mathrm{x}}\right)^5 = -{}^{10}\mathrm{C}_5 \frac{1}{2^5} = \frac{-63}{8}$

3. [1] Hint : use $r = \frac{np}{p+q}, n = 9, p = 2, q = 1$

4. [3] Sol: $\mathrm{T}_{\mathrm{r}+1} = {}^{10}\mathrm{C}_\mathrm{r}\left(\frac{\mathrm{x}}{2}\right)^{10-\mathrm{r}} \cdot \left(\frac{-3}{\mathrm{x}^2}\right)^{\mathrm{r}}$

$= {}^{10}C_r\left(\frac{1}{2}\right)^{10-r} \cdot (-3)^r \frac{x^{10-r}}{x^{2r}} = {}^{10}C_r\left(\frac{1}{2}\right)^{10-r} \cdot (-3)^r x^{10-3r}$

To get the coefficient of x^4 equate x power to 4 .

$10 - 3r = 4 \Rightarrow 3r = 6 \Rightarrow r = 3$

Coefficient of x^4 is ${}^{10}C_2\left(\frac{1}{2}\right)^8 \cdot (-3)^2 \Rightarrow \frac{405}{256}$

5. [4] rth term from the ending is $(n - r + 2)$ th term from starting.

$\mathrm{T}_{8-3+2} = \mathrm{T}_7 = \mathrm{T}_{6+1} = {}^8\mathrm{C}_6\left(\frac{3\mathrm{x}}{5}\right)^2\left(\frac{-5}{2\mathrm{x}}\right)^6$

$= {}^8\mathrm{C}_2\left(\frac{3}{5}\right)^2\left(\frac{-5}{2}\right)^6\frac{\mathrm{x}^2}{\mathrm{x}^6} = \frac{8\times7}{2\times1}\cdot\frac{3^2}{5^2}\frac{(-5)^6}{2^6}\cdot\frac{1}{\mathrm{x}^4} = \frac{39375}{16\mathrm{x}^4}$

6. [3] Use the formula $\frac{(n+1)|x|}{1+|x|}$ 7. [1]Sol: In $(a+b+c)^{18}$, general term is given by $\frac{18!}{(r_1)!(r_2)!(r_3)!}(a)^{r_1}(b)^{r_2}(c)^{r_3}$

Coefficient of $a^8b^6c^4$ is obtained by putting $r_1 = 8, r_2 = 6, r_3 = 4$ in (1)

i.e., $\frac{(18)!}{8!.6!.4!}a^8b^6c^4$. Thus required coefficient $= \frac{(18)!}{8!.6!.4!} \times \frac{14!}{14!} = {}^{18}C_{14} \times {}^{14}C_8$

8. [21]Coefficient of T_r in $(3+7\mathrm{x})^{29}$ is ${}^{29}\mathrm{C}_{\mathrm{r}-1}3^{29-(r-1)}7^{r-1} = {}^{29}\mathrm{C}_{\mathrm{r}-1}3^{30-r}7^{r-1}$
and Coefficient of $\mathrm{T}_{\mathrm{r}+1}$ in $(3+7\mathrm{x})^{29}$ is ${}^{29}\mathrm{C}_\mathrm{r}3^{29-\mathrm{r}}7^\mathrm{r}$

Given $^{29}C_{r-1}3^{30-r}7^{r-1} = {}^{29}C_r 3^{29-r}7^r \Rightarrow r = 21$

9. [3] Sol: We know that $(x-1)^6 = x^6 - 6x^5 + 15x^4 - 20x^3 + 15x^2 - 6x + 1$
$(x+1)^6 = x^6 + 6x^5 + 15x^4 + 20x^3 + 15x^2 + 6x + 1$

$(x-1)^6 + (x+1)^6 = 2[x^6 + 15x^4 + 15x^2 + 1] = 2x^6 + 30x^4 + 30x^2 + 2$

10. [4] $\frac{f(1)-f(-1)}{2}$

11. [3] Given coefficient of x in $\left(x^2 + \frac{k}{x}\right)^5$ is 270

$\Rightarrow T_{r+1} = {}^5C_r(x^2)^{5-r}k^r x^{-r} = {}^5C_r k^r x^{10-3r}$

To get the coefficient of $x, 10 - 3r = 1 \Rightarrow r = 3$

$T_4 = {}^5C_3 k^3 x = 270x \Rightarrow k^3 = 27 \Rightarrow k = 3$

$2^{2000} = (2^4)^{500} = 16^{500} = (17-1)^{500}$

12. [1] $(17-1)^{500} = {}^{500}C_0 17^{500} - {}^{500}C_1 17^{499} + \cdots \ldots \ldots - {}^{500}C_{499}17 + 1$
$= 17k + 1$

2^{2000} leaves remainder when divided by 17 is 1

13. [3] $T_3 = 10^6 \Rightarrow 5_{C_2} x^3 \left(x^{\log_{10}^x}\right)^2 = 10^6, x = 10$

14. [3] General term in the expansion of $(\sqrt[4]{5} + \sqrt[5]{4})^{100}$ is $T_{r+1} = {}^{100}C_r (5)^{\frac{100-r}{4}} \left(4^{\frac{1}{5}}\right)^r$

To get the rational terms $\frac{100-r}{4}$ has to be integer so the possible values of $r =$ 0,4,8,16,32, …… ,100

and also $r/5$ is also integer so the possible values of r = 0,5,10,15,20 …,100 Hence the possible values in both cases are 0,20,40,60,80 and 100 .

15. [1] $(n+1)(n+2) = 72$

16. [1] Coefficient of T_{2r+1} = Coefficient of T_{4r+5} in $(1+x)^{10}$

$$^{10}C_{2r} = {}^{10}C_{4r+4}$$
$$\frac{10!}{(10-2r)!\,(2r)!} = \frac{10!}{(10-4r-4)!\,(4r+4)!}$$

$\Rightarrow 2r = 4r + 4$ or $2r + 4r + 4 = 10$
$\Rightarrow r = 1$ or $r = 1$

17. [1] Coefficient of x^5 in $(1+x)^{10}\left(1+\frac{1}{x}\right)^{20}$ = Coefficient of x^{25} in $(1+x)^{30}$

18. [4] $(2+\sqrt{3})^n = [x] + f$. Let$(2-\sqrt{3})^n = f^1, 0 < f^1 < 1$

$[x] + f + f^1 = 2[\,{}^nC_0 2^n + {}^nC_2 2^{n-2}3 + \cdots]$ an even integer
$\therefore f + f^1 = 1 \;\therefore [x]$ is an odd integer , $f^1 = 1 - f$

But $([x]+f)f^1=(2+\sqrt{3})^n(2-\sqrt{3})^n=1$

$\therefore [x]=\frac{1}{f^1}-f=\frac{1}{1-f}-f=\frac{1-f+f^2}{1-f}\ \ \therefore \frac{f^2}{1-f}=[x]-1$, an even integer

19. [7] use multinomial theorem.

General term of given expansion is $\frac{6!}{a!b!c!}1^a 2^{\frac{b}{2}}3^{\frac{c}{3}}$ where $a+b+c=6$ and

20. [1]

a	b	c
6	0	0
4	2	0
2	4	0
0	6	0
3	0	3
1	2	3
0	0	6

21. [1] Given that $\boldsymbol{I}+\boldsymbol{f}=(2+\sqrt{3})^{\boldsymbol{n}};0\le \boldsymbol{f}<1\ \boldsymbol{F}=(2-\sqrt{3})^{\boldsymbol{n}},0<\boldsymbol{F}<1$

$\boldsymbol{I}+\boldsymbol{f}+\boldsymbol{F}=2[{}^n\boldsymbol{C}_6 2^n+{}^n\boldsymbol{C}_2 2^{n-2}\cdot 3+\cdots..]=2\times$ integer

since $\boldsymbol{o}<\boldsymbol{f}<\boldsymbol{F}<2\Rightarrow \boldsymbol{f}+\boldsymbol{F}=\boldsymbol{I}$ i.e $\boldsymbol{F}=\boldsymbol{I}-\boldsymbol{f}$

$(\boldsymbol{I}+\boldsymbol{f})(\boldsymbol{I}-\boldsymbol{f})=(\boldsymbol{I}+\boldsymbol{f})\boldsymbol{F}=(2+\sqrt{3})^n(2-\sqrt{3})^n=1$

Level – III

22. B, C, D	23.B	24.A	25.B	26.a-r;b-p;c-s;d-q
27.5	28.A	29.B	30.C	31. C, D 32. A, B, C, D
33.A	34.2	35.2		

22. [2,3,4]

23. [2] Sol: $(x+a)^n=x^n+{}^nC_1x^{n-1}a+{}^nC_2x^{n-2}+{}^nC_3x^{n-3}a^3+\cdots..+{}^nC_na^n$

$=(x^n+{}^nC_2x^{n-2}a^2+\cdots.+({}^nC_1x^{n-1}a+{}^nC_3x^{n-3}a^3+\cdots)=P+Q$

(1) and $(x-a)^n=x^n-{}^nC_1x^{n-1}a+{}^nC_2x^{n-2}a^2-{}^nC_3x^{n-3}a^3+\cdots...+(-1)^n{}^nC_na^n$

$=(x^n+{}^nC_2x^{n-2}a^2+\cdots...)({}^nC_1x^{n-1}a+{}^nC_3x^{n-3}a^3+\cdots...)=P-Q.$

$P^2-Q^2=(P+Q)(P-Q)=(x+a)^n(x-a)^n=(x^2-a^2)^n$

24. [1] Sol: $4PQ=(P+Q)^2-(P-Q)^2=(x+a)^{2n}-(x-a)^{2n}$ 25. [2] Sol: $P^2+Q^2=(P+Q)^2-2PQ$

25. a-r; b-p;c-s; d-q

a) Since $n=10$ is an even number. Then the given expansion has only one middle term. i.e., $\frac{T_n}{2}+1=T_{5+1}=T_6;T_6=10_{c5}\left(\frac{3x}{7}\right)^{10-5}\cdot(-2y)^5=-10_{c5}\left(\frac{6}{7}\right)^5\cdot x^5\cdot y^5$ b) Since $n=8$ is an even number. Then the given expansion has only one middle term

i.e., $T_5=8_{c4}(7p)^4\cdot(2q)^4$

c) Since n = 17 is an odd number. Then the expansion has two middle terms i.e., T_9& T_{10}

$T_9 = 17_{c8}(4x^2)^{17-8} \cdot (5x^3)^8 = 17_{c8}4^9 \cdot 5^8 \cdot x^{42}$;

$T_{10} = 17_{c9}(4x^2)^{17-9} \cdot (5x^3)^9 = 17_{c9}4^8 \cdot 5^9 \cdot x^{43}$

d) Since $n = 20$ is even number. Then the expansion has only one middle term i.e., $T_{11} = 20_{c_{10}}\left(\frac{3}{a^3}\right)^{20-10} \cdot (5a^4)10 = 20_{c_{10}}(15)^{10} \cdot a^{10}$

27. Ans : 5; Since $n = 9$ is odd the expansion of $(1 + 5\sqrt{2}x)^9 + (1 - 5\sqrt{2}x)^9$ has $\left(\frac{n+1}{2}\right)$ i.e., $\frac{9+1}{2} = 5$ terms

28. Here n = 19, an odd integer. ⇒∴ The largest binomial coefficients are $^{19}C_9$& $^{19}C_{10}$.

29. [2] Here n = 24 is an even integer

∴ The largest binomial coefficient is $^{24}C_{12}$.

30. [3] Here n = 22 is an even integer. ∴ The largest binomial coefficient is $n_{c_{\left(\frac{n}{2}\right)}} = 22_{c_{11}} = 22_{c_r} \Rightarrow r = 11;\ 13_{c_r} = 13_{c_{11}} = 13_{c_2} = \frac{13\times12}{1\times2} = 78$

31. [3,4] Given

$T_5 = 24\ T_3$

$^{11}C_4x^4 = 24\ ^{11}C_2x^2$

$$\frac{11!}{(11-4)!\,4!}x^2 = 24\frac{11!}{(11-2)!\,2!}$$

$x^2 = 4$

$x = \pm 2$

32. [1,2,3,4] Sum of coefficients in $(a\alpha^2x^2 + 2b\alpha x + c)^n$ is $(a\alpha^2 + 2b\alpha + c)^n$.
Let $f(\alpha) = a\alpha^2 + 2b\alpha + c$

Such that sum = $[f(\alpha)]^n$. Now, $f(\alpha) > 0\forall n$, and $\alpha \in R$ then $a > 0$. Also, $f(\alpha) < 0\forall n \in$ odd and $\alpha \in R$ if $a < 0$. Similarly, options (b) and (d) are also true.

33. [1] Sol: $T_2 = 135, T_3 = 30, T_4 = 10/3$
$n_{c_1} \cdot a^{n-1} \cdot b^1 = 135 \rightarrow 1;\ n_{c_2}a^{n-2} \cdot b^2 = 30 \rightarrow 2;\ n_{c_3}a^{n-2} \cdot b^3 = 10/3 \rightarrow 3$

$$\Rightarrow \frac{n_{c_2} \cdot a^{n-2}b^2}{n_{c_1}a^{n-1}b} = \frac{30}{135}; \frac{(n-1)(a^{-1}) \cdot b}{2} = \frac{2}{9};\ 9(n-1)b = 4a \rightarrow (4) \Rightarrow \frac{n_{c_3}a^{n-3}b^3}{n_{c_2}a^{n-2}b^2} = \frac{\frac{10}{3}}{30}$$

$(n-2)(a^{-1})b = \frac{1}{3};\ 3\ b(n-2) = a \rightarrow (5) \Rightarrow \frac{(4)}{(5)} \Rightarrow 3(n-1) = 4(n-2)$

$4n - 3n - 8 + 3 = 0 \Rightarrow n = 5$

34. [2] Sol: $T_{21} = 44_{c_{20}} \cdot x^{20}$; $T_{22} = 44_{c_{21}} \cdot x^{21}$. Given that $T_{21} = T_{22}$

$\Rightarrow 44_{c_{20}} x^{20} = 44_{c_{21}} x^{21}$

$\Rightarrow \frac{44!}{(24!)20!} = \frac{44!}{(23)!\,(21)!} \cdot x \Rightarrow \frac{21}{24} = x \Rightarrow x = 7/8$

35. [2] Sol: The rth term in the given expansion is $T_r = 18_{c_{(r-1)}} \cdot x^{r-1}$ Thus the coefficient of rth term $= 18_{c_{r-1}}$

WORKSHEET-2

Level – I

1) 2 2) 4 3) 4 4. 3 Hint: $n_{C_r} = \frac{n!}{(n-r)!r!}$ 4.3

5.3 6.1 7.1

Sol:

$C_0 + C_2 + C_4 + C_8 = 2^{9-1} = 2^8 = 256$

8. $3\sum_{r=1}^{n} r \cdot {}^{n}C_r = \sum_{r=1}^{n} r \cdot \frac{n}{r} {}^{n-1}C_{r-1} = \sum_{r=1}^{n} n \cdot {}^{n-1}C_{r-1} = n\{{}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \cdots + {}^{n-1}C_{n-1}\} = n \cdot 2^{n-1}$

9. 4 10.2

Level – II

1.D	2.A	3.D	4.D	5.D	6.C	7.B
8.A	9.2	10.D	11.D	12.A	13.8	14.B
15.C	16.A	17.A	18.C	19.4	20.A	21.D

1. [4] In the expansion of $(1+x)^{2n}$ sum of coefficients is

${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \cdots + {}^{2n}C_{2n} = 2^{2n}$

${}^{2n}C_0 + {}^{2n}C_1 + {}^{2n}C_2 + \cdots + {}^{2n}C_{2n} = 2^{2n-1} \therefore {}^{2n}C_2 + {}^{2n}C_4 + \cdots + {}^{2n}C_{2n} = 2^{2n-1} - 1$ 2. [1] sum of coefficients ${}^{n}C_0 + {}^{n}C_1 + \cdots + {}^{n}C_n = 2^n$;

${}^{n}C_1 + {}^{n}C_3 + \cdots + {}^{n}C_{n-1} = 2^{n-1}$

3. [4] $C_0 + 2C_1 + 3C_2 + \cdots + (n+1)C_n$

$$\sum_{r=0}^{n} (r+1)C_r = \sum_{r=0}^{n} r \cdot C_r + \sum_{r=0}^{n} C_r = n2^{n-1} + 2^n = n \cdot 2^{n-1} + 2 \cdot 2^{n-1} = (n+2)2^{n-1}$$

4. [4] $\frac{C_0}{2} - \frac{C_1}{3} + \frac{C_2}{4} - \frac{C_3}{5} + \cdots$

$$= \frac{1}{(n+1)(n+2)}\left[\frac{(n+2)(n+1)}{2!} - \frac{2 \cdot (n+2)(n+1) \cdot n}{3!} + \frac{3 \cdot (n+2)(n+1)n(n-1)}{4!} - \cdots\right]$$

$$= \frac{1}{(n+1)(n+2)}\left[{}^{(n+2)}C_2 - {}^{2(n+2)}C_3 + {}^{3(n+2)}C_4 - \cdots..\right]$$

5. [4] $C_0^2 + C_1^2 + C_2^2 + \cdots \ldots \ldots .. + C_n^2 = {}^{2n}C_n$

6. [3] Standard Result

7. [2] Range of $(4+5x)^{-\frac{7}{3}} = 4^{-\frac{7}{3}}\left(1+\frac{5x}{4}\right)^{-\frac{7}{3}}$ is

$$\left|\frac{5x}{4}\right| < 1 \Rightarrow \frac{-4}{5} < x < \frac{4}{5}$$
$$x \in \left(\frac{-4}{5}, \frac{4}{5}\right)$$

8. [1] In $(1+2x)^{\frac{1}{2}}$ General term $= \frac{n(n-1)(n-2)\ldots\ldots(n-(r-1))}{r!}x^r$

$$= \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\ldots\ldots\left(\frac{1}{2}-(r-1)\right)}{r!}x^r$$
$$= \frac{\left(\frac{1}{2}\right)\left(\frac{-3}{2}\right)\left(\frac{-5}{2}\right)\ldots\ldots\left(\frac{2r-3}{2}\right)}{r!}(-1)^{r-1}2^r x^r = \frac{1.3.5\ldots(2r-3)}{r!}x^r$$

9. $\sum(-1)^{r-1}(r+2)C_r = \sum(-1)^{r-1}r^n C_r + \sum(-1)^{r-1}2C_r$

$$= \sum\ (-1)^{r-1}n^{(n-1)}C_{r-1} + 2\cdot\sum\ (-1)^{r-1}C_r = 0 - 2[0-1] = 2$$

10. [4] ${}^nC_r + {}^nC_{r-1} = {}^{(n+1)}C_r$

11. [4] Given expression contains $3n$ factors Choose r brackets out of $3n$ brackets to get a term of degree r. Hence the coefficient is ${}^{3n}C_r$. Let $n = 3$ verify 12. [1] $C_0 - C_1x + C_2x^2 - \cdots . = (1-x)^n$; $C_0x - C_1x^2 + C_2x^3 - \cdots = x(1-x)^n$

Differenation yields $1.C_0 - 2C_1x + 3\cdot C_2x^2 - \cdots . = (1-\overline{n+1}x)(1-x)^{n-1}$ multiply with x then differentiate and substiute $x = 1$

13. [8] $(1+x)^{\frac{27}{5}} = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$

$$+\frac{\mathrm{n(n-1)(n-2)(n-3)(n-4)(n-5)(n-6)}}{7!}\mathrm{x}^7 + \cdots.$$

For $n = \frac{27}{5}$ upto $n(n-1)(n-2)(n-3)(n-4)(n-5)$ give positive value remaining are negative hence for $r = 6$ is the first negetive term. Hence 7th term is first negative term.

14. [2] assign n value

15. [3] General term is $\frac{n(n+1)(n+2)(n+3)\ldots\ldots\ldots(n+r-1)}{r!}x^r$

$$\frac{1}{\sqrt{2a}}\left(\frac{\frac{1}{2}\left(\frac{1}{2}+1\right)\left(\frac{1}{2}+2\right)\ldots\ldots\left(\frac{1}{2}+r-1\right)}{r!}\right)\left(\frac{3b}{2a}\right)^r$$
$$= \frac{1}{\sqrt{2a}}\left(\frac{1.3.5\ldots\ldots(2r-1)}{2^r r!}\right)\left(\frac{3b}{2a}\right)^r = \frac{1}{\sqrt{2a}}\left(\frac{1.3.5\ldots\ldots(2r-1)}{r!}\right)\left(\frac{3b}{4a}\right)^r$$

16. [1] $\frac{(1+3x)^{\frac{1}{2}}(1+x)^{\frac{2}{3}}}{(1-x)^{\frac{1}{2}}} = (1+3x)^{\frac{1}{2}}(1+x)^{\frac{2}{3}}(1-x)^{-\frac{1}{2}}$

$= \left(1-\frac{1}{2}\cdot 3x\right)\left(1+\frac{2}{3}\cdot 3x\right)\left(1+\frac{1}{2}x\right)$

$= 1-\frac{3x}{2}+\frac{2x}{3}+\frac{x}{2} = 1-\frac{9x+4x+3x}{6} = 1-\frac{x}{3}$

17. [1] $\sum_{r=0}^{n-1}\left(\frac{{}^{n}C_r}{{}^{n}C_r+{}^{n}C_{r+1}}\right)^3 = \sum_{r=0}^{n-1}\left(\frac{{}^{n}C_r}{{}^{n}C_r\left(1+\frac{n-r}{r+1}\right)}\right)^3 = \sum_{=0}^{n-1}\left(\frac{r+1}{n+1}\right)^3 = \frac{n^2}{4(n+1)} = \frac{4}{5}$ gives $n = 4$

18. [3] $(1+x)^2(8-x)^{\frac{-1}{3}} = (1+2x+x^2)8^{\frac{-1}{3}}\left(1-\frac{x}{8}\right)^{\frac{-1}{3}}$

Put $r = 2$ to get the x^2 coefficient

$= \frac{1}{2}\left(1+\frac{1}{288}+\frac{1}{24}\cdot 2+1\cdot 1\right) = \frac{1}{576}+\frac{1}{24}+\frac{1}{2} = \frac{313}{576}$ 19. [4] ${}^{95}C_4 + {}^{95}C_3 + {}^{96}C_3 + {}^{97}C_3 + {}^{98}C_3 + {}^{99}C_3$

$= {}^{96}C_4 + {}^{96}C_3 + {}^{97}C_3 + {}^{98}C_3 + {}^{99}C_3 = {}^{97}C_4 + {}^{97}C_3 + {}^{98}C_3 + {}^{99}C_3$
$= {}^{98}C_4 + {}^{98}C_3 + {}^{99}C_3 = {}^{99}C_4 + {}^{99}C_3 = {}^{100}C_4$

20. [1] The given series is equal to $\frac{1}{2}\int((1+x)^{50}+(1-x)^{50})dx$

21. [4] $(1+x)^n = n_{C_0} + n_{C_1}x + n_{C_2}x^2 + \cdots \ldots + n_{C_n}x^n$

$x(1+x)^n = n_{C_0}x + n_{C_1}x^2 + n_{C_2}x^3 + \cdots \ldots + n_{C_n}x^{n+1}$

Put $x = 1, \omega, \omega^2$ and add $\Rightarrow C_2 + C_5 + C_8 + \cdots \ldots = \frac{1}{3}\{2^n + (-1)^n(\omega^{2n+1} + \omega^{n+2})\}$

$= \frac{1}{3}\{2^{2014} + (-1)^{2014}(\omega^{4029} + \omega^{2016})\} = \frac{2^{2014}+2}{3}$

Level – III

22.A,B	23.A	24.B	25.C	26.A	27.a-p;b-r;c-q;d-s
28.0	29. B, C	30.a-p;b-q;c-r;d-s		31.8	32.A,B,C
33.A	34.D	35.D			

22. (1,2) Sol: $k = \frac{(C_0+C_1)(C_1+C_2)\ldots(C_{n+1}+C_n)}{C_0C_1\ldots C_0} = \frac{{}^{(n+1)}C_1\,{}^{(n+1)}C_2\ldots{}^{(n+1)}C_n}{{}^{n}C_0\,{}^{n}C_1\ldots{}^{n}C_n} = \frac{(n+1)^n}{n!}$

23. (1)

24. (2) Sol: L.H.S $= {}^{15}C_2 + 2\,{}^{15}C_3 + \cdots . + 14\cdot{}^{15}C_{15}$

25. (3)

$= 1 + (-1)\cdot{}^{15}C_0 + 0\cdot{}^{15}C_1 + 1\cdot{}^{15}C_2 + 2\cdot{}^{15}C_3 + \cdots + 14\cdot{}^{15}C_{15} = 1 + (-1+14)\cdot 2^{15-1}$
$= 13\cdot 2^{14} + 1$

Sol: $c_0 + C_1 + c_2 + \cdots + C_n = 2^n = 128 \Rightarrow n = 7$. Now

$C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \frac{C_3}{4} + \cdots . (-1)^n \frac{C_n}{n+1} = \frac{1}{1+7} = \frac{1}{8}$

26. (1) Sol: $2 \cdot c_0 + 7c_1 + 12 \cdot c_2 + \cdots + (5n+2) \cdot c_n = (2a + nd) \cdot 2^{n-1} = (4 + 5n)2^{n-1}$

27. Key : a-p; b-r; c-q; d-s

Sol: a) Given $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \cdots . + a_{2n} x^{2n}$

Put $x = 1$, then, $3^n = a_0 + a_1 + a_2 + \cdots + a_{2n}$

b) Then $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \cdots . + a_{2n} x^{2n}$

Put $x = 1$, then $3^n = a_0 + a_1 + a_2 + \cdots . + a_{2n} \rightarrow (1)$

Put $x = -1$, then $1 = a_0 - a_1 + a_2 \ldots + a_{2n} \rightarrow (2)$

adding (1) and (2) equations.

$\Rightarrow 3^n + 1 = (a_0 + a_1 + a_2 + \cdots . + a_{2n}) + (a_0 - a_1 + a_2 + \cdots . + a_{2n})$

$\Rightarrow (2a_0 + 2a_2 + \cdots . + 2a_{2n}) = 3^n + 1 \Rightarrow a_0 + a_2 + \cdots . + a_{2n} = \frac{3^n + 1}{2}$

c) Subtracting above (1) and (2) equations

$\Rightarrow 3^n - 1 = (a_0 + a_1 + a_2 + \cdots . + a_{2n}) - (a_0 - a_1 + a_2 + \cdots . + a_{2n}) \Rightarrow a_1 + a_3 + \cdots . + a_{2n-1} = \frac{3^n - 1}{2}$

d) Given expansion $(1 + x + x^2)^n = a_0 + a_1 x + a_2 x + \cdots . + a_{2n} x^{\text{en}}$

Put $x = 1$, then $3^n = a_0 + a_1 + \cdots + a_{2n} \rightarrow (1)$

Put $x = \omega$, then, $(1 + \omega + \omega^2) = 0 = a_0 + a_1\omega + a_2\omega^2 + \cdots + a_{2n}(\omega)^{2n} \rightarrow (2)$

Put $x = \omega^2$, then, $a_0 + a_1\omega^2 + a_2\omega^4 + \cdots + a_{2n}\omega^{4n} = 1 + \omega^2 + \omega^4 = 0 \rightarrow (3)$

adding (1), (2) and (3)

$3a_0 + a_1(1 + \omega^2 + \omega^4) + a_3(1 + \omega^3 + \omega^6) + \cdots + a_{2n}(1 + \omega^{2n} + \omega^{4n}) = 3^n$

28. Ans : 0 Sol. sum $= \sum(-1)^n \cdot (n+2) \cdot C_n = \sum(-1)^n \cdot {}^nC_n + 2\sum(-1)^n \cdot C_n = 0 + 0 = 0$

29. (2,3)

30. a-p; b-q; c-r; d-s

31. [8] $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \cdots \infty$

So $(1 - 2x + 3x^2 - 4x^3 + \cdots \infty)^{-4} = ((1 + x)^{-2})^{-4} = (1 + x)^8$

Coefficient of x^7 in $(1 + x)^8$ is ${}^8C_7 = 8$

32. [1,2,3] $S = C_0^2 + \sum_{r=1}^{n} \frac{C_r^2}{C_{r-1}} = 1 + \sum[(n+1)C_r - {}^rC_r] = 1 + (n+1)(2^n - 1) - n \cdot 2^{n-1}$

$= 1 + n \cdot 2^n + 2^n - -n \cdot 2^{n-1} = n \cdot 2^{n-1} + 2^n - n\,S + n = 2^{n-1}(n+2)$

33. [1] $\sum_{0\le i}\sum_{<j\le 50} a_i a_j = \frac{1}{2}[2^{50}\cdot 2^{50} - {}^{100}C_{50}] = 2^{99} - {}^{99}C_{50}$

34. [4] $\sum_{0\le i<j\le 50}\sum_i (i+j)a_i a_j = \frac{1}{2}\left[\sum_{i=0}^{50}\sum_{j=0}^{50}(i+j)^{50}C_i^{50}C_j - \sum_{i=0}^{50}2i^{50}C_i^2\right]$

$$= \frac{1}{2}\left[\sum_{i=0}^{50}\sum_{j=0}^{50} i^{50}C_i^{50}C_j + \sum_{i=0}^{50}\sum_{j=0}^{50} j^{50}C_j^{50}C_i - \sum_{i=0}^{50}U^{50}C_0^2\right]$$

$$= \frac{1}{2}\left[2^{50}\cdot\sum_{i=1}^{50} i^{50}C_i + 50.2^{49}\sum_{i=0}^{50}{}^{50}C_i - 2\cdot\sum_{i=0}^{50}5\cdot{}^{49}C_i^{50}C_i\right]$$

$$= \frac{1}{2}[2^{50}, 50\cdot 2^{49} + 50\cdot 2^{49}\cdot 2^{50} - 250\cdot{}^{99}C_{50}] = 50(2^{99} - {}^{99}C_{50})$$

35. [4] $\sum_{0\le i}\sum_{j\le 50}(a_i - a_j)^2 = \frac{1}{2}\sum_{i=0}^{50}(a_i^2 + a_j^2 - 2a_i a_j) = \frac{1}{2}\left[50\sum_{i=0}^{50}90^2 + \sum_{i=0}^{50}{}^{100}C_{50} - 2\cdot 2^{50}\cdot 2^{50}\right]$

$$= \frac{1}{2}[51\cdot{}^{100}C_{50} + 51\cdot{}^{100}C_{50} - 2.2^{100}] = 51.{}^{100}C_{50} - 2^{100} = 102.{}^{99}C_{49} - 2^{100}$$

2. DIFFERENTIATION

WORKSHEET-1

Level – I

1. A	2. B	3. C	4. C	5. C	6. A	7. A
8. B	9. B	10. A	11. D			

Level – II

1. B	2. C	3. B	4. A	5. B	6. B	7. 0
8. A	9. A	10. A	11. A	12. 2	13. A	14. B
15. 4	16. A	17. B	18. A	19. C	20. 0	

1. $\frac{d}{dx}\left[3\sqrt[3]{x}-\frac{4}{\sqrt[4]{x}}\right]=3\frac{1}{3}x^{-\frac{2}{3}}-4\left(-\frac{1}{4}\right)x^{\frac{-5}{4}}\Rightarrow x^{\frac{-2}{3}}+x^{\frac{-5}{4}}$

2. $y=3^{\log_9(1+\tan^2 x)}\Rightarrow y=3^{\log_3\sec^2 x}=\sec x\Rightarrow\frac{dy}{dx}=\sec x\tan x$

3. $\frac{d}{dx}\left(\frac{a-b\cos x}{a+b\cos x}\right)=\frac{(a+b\cos x)(b\sin x)-(a-b\cos x)(-b\sin x)}{(a+b\cos x)^2}=\frac{2ab\sin x}{(a+b\cos x)^2}$

4. $y=\cos(1-x^2)^2\Rightarrow\frac{dy}{dx}=-\sin(1-x^2)^2\cdot 2(1-x^2)\cdot(-2x)=-4x(1-x^2)\sin(1-x^2)^2$

5. $y=\log\frac{1-x^2}{1+x^2}\Rightarrow y=\log(1-x^2)-\log(1+x^2)\Rightarrow\frac{dy}{dx}=\frac{1}{1-x^2}(-2x)-\frac{1}{1+x^2}(2x)$

6. $f^1(x)=\begin{cases}1,0\le x\le 2\\0,\ x\ge 2\end{cases}\Rightarrow f^1(2^+)=0, f^1(2^-)=1\Rightarrow f^1(2^+)\ne f^1(2^-)$

7. $f(-x)=f(x)\Rightarrow f^1(x)+f^1(-x)=0\Rightarrow f^1(0)+f^1(0)=0\Rightarrow f^1(0)=0$

8. $f(x)=\begin{cases}x^2,x>0\\-x^2,x<0\end{cases}\Rightarrow f'(x)=\begin{cases}2x,x>0\\-2x,x<0\end{cases}\Rightarrow f'(0^+)=f'(0^-)$

9. $y=(\log x)^3\Rightarrow\frac{dy}{dx}=3(\log x)^2\frac{1}{x}=\frac{3(\log x)^2}{x}$

10. $\frac{d}{dx}(\sqrt{\sin\sqrt{x}})=\frac{1}{2\sqrt{\sin\sqrt{x}}}\frac{d}{dx}(\sin\sqrt{x})=\frac{\cos\sqrt{x}}{2\sqrt{\sin\sqrt{x}}}\frac{1}{2\sqrt{x}}=\frac{\cos\sqrt{x}}{4\sqrt{x\sin\sqrt{x}}}$

11. $\frac{d}{dx}(\sin(\sqrt{2x-3}))=\cos\sqrt{2x-3}\frac{1}{2\sqrt{2x-3}}\cdot 2=\frac{\cos\sqrt{2x-3}}{\sqrt{2x-3}}$

12. $y=2^{\mathrm{ax}}\Rightarrow\frac{dy}{dx}=2^{ax}\cdot\log 2\cdot a\Rightarrow\log 256=a\cdot 2^{ax}\cdot\log 2\Rightarrow\log 2^8=a\cdot 2^{ax}\cdot\log 2$

$\Rightarrow 8\log 2=a\cdot 2^{\mathrm{ax}}\log 2\ 8=\mathrm{a}\cdot 2^{\mathrm{a}}\Rightarrow \mathrm{a}=2$

13. $\frac{d}{dx}(\log(\log(\log x)))=\frac{1}{\log(\log x)}\frac{1}{\log x}\cdot\frac{1}{x}$

14. $f(x)=x-x^2+x^3-x^4+\cdots\ldots\propto$

$$f(x)=\frac{x}{1+x}\Rightarrow f'(x)=\frac{1}{(1+x)^2}\quad\left[r=\frac{-x^2}{x}=-x,|r|<1\ \mathrm{S}_\infty=\frac{a}{1-r}\right]$$

15. $\lim_{x\to 2}\frac{xf(x)-2f(x)}{x-2}$

$$=\lim_{x\to 2}\frac{xf^1(x)+f(x)-2f^1(x)}{1}=2f^1(2)+f(2)-2f^1(2)=2(1)+4-2(1)=4$$

16. $\frac{d}{dx}\left(\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right)=\frac{x}{2}\frac{1}{2\sqrt{a^2-x^2}}(-2x)+\sqrt{a^2-x^2}\cdot\frac{1}{2}+\frac{a^2}{2}\frac{1}{\sqrt{a^2-x^2}}$

$=\frac{-x^2+a^2-x^2+a^2}{2\sqrt{a^2-x^2}}=\frac{2(a^2-x^2)}{2\sqrt{a^2-x^2}}=\sqrt{a^2-x^2}$

17. Put $x=\sin\alpha, y=\sin\beta \Rightarrow \cos\alpha+\cos\beta=a(\sin\alpha-\sin\beta)$

$2\cos\frac{(\alpha+\beta)}{2}\cdot\cos\left(\frac{\alpha-\beta}{2}\right)=\mathrm{a}\left(2\cos\frac{\alpha+\beta}{2}\cdot\sin\frac{\alpha-\beta}{2}\right)$

$\frac{\alpha-\beta}{2}=\cot^{-1}a\Rightarrow\alpha-\beta=2\cot^{-1}a\Rightarrow\sin^{-1}x-\sin^{-1}y=2\cot^{-1}a$

$\Rightarrow\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0\Rightarrow\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$

18. $\sqrt{\tan y}=e^{\cos 2x}\cdot\sin x$

$\log(\tan y)^{1/2}=\log e^{\cos 2x}\cdot\sin x\Rightarrow\frac{1}{2}\log(\tan y)=\cos 2x\log_e^e+\log\sin x$

$\Rightarrow\frac{1}{2}\cdot\frac{1}{\tan y}\cdot\sec^2 y\frac{dy}{dx}=-\sin 2x\cdot 2+\frac{1}{\sin x}\cdot\cos x$

$\Rightarrow\frac{1}{2}\frac{1}{\mathrm{Tan}\,y}\sec^2 y\frac{dy}{dx}=\cot x-2\sin 2x\Rightarrow\frac{1}{\sin 2y}\frac{dy}{dx}=\cot x-2\sin 2x$

$\Rightarrow\frac{dy}{dx}=\sin 2y(\cot x-2\sin 2x).$

19. $\frac{d}{dx}\left(\frac{1+x^2+x^4}{1+x+x^2}\right)=ax+b\Rightarrow\frac{d}{dx}\frac{\left((x^2+1)-x\right)(x^2+1+x)}{(x^2+x+1)}=ax+b.$

$\Rightarrow\frac{d}{dx}(x^2+1-x)=ax+b\Rightarrow 2x-1=ax+b\Rightarrow a=2,b=-1$

20. $f(x)=|x-1|+|x-3|\Rightarrow f(x)=x-1-(x-3)$ Since, $1<x<3$

$=x-1-x+3\Rightarrow f(x)=2\Rightarrow f^1(x)=0.$

Level – III

21. A,B	22. A	23. C	24. D	25. C	26.a-r; b-q; c-p; d-s,t
27. 2	28. A,B,C,D	29. B	30. A	31. B	32. a-r; b-s; c-q; d-p
33. 0	34. A,B,C	35. 1			

21. Conceptual

22. Conceptual 23. $f^1(x)=10$ 24. $D[x^2]=2x.$

25. $f(x)=k\Rightarrow f^1(x)=0.$

26. a) $\frac{d}{dx}(2x+1)=2$

b) $\frac{d}{dx}(3x^2)=6x$

c) $\frac{d}{dx}(5)=0$

d) $\frac{d}{dx}[f(x)]=f^1(x)=Lt\frac{f(x+h)-f(x)}{h}$

27. $\mathrm{Lt}_{h\to 0}\frac{2010(x+h)-2010x}{h}=2010\Rightarrow\frac{2010}{1005}=2$

28. Conceptual.

29. $f(x) = \log(x^2+1) \Rightarrow f^1(x) = \frac{1}{x^2+1}(2x)$.

30. $f(x) = \sin(\sqrt{2x-3}) \Rightarrow f^1(x) = \cos\sqrt{2x-3} \cdot \frac{1}{2\sqrt{2x-3}} \cdot 2 = \frac{\cos\sqrt{2x-3}}{\sqrt{2x-3}}$

31. $\frac{d}{dx}\left(\frac{\sin x}{1+\cos x}\right) = \frac{d}{dx}\left(\frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right) = \frac{d}{dx}\text{Tan}\frac{x}{2} = \sec^2\frac{x}{2} \cdot \frac{1}{2} = \frac{1}{2\cos^2\frac{x}{2}} = \frac{1}{1+\cos x}$

32. $f(x) = 2^x$

$g(x) = 3^x$

$f^1(x) = 2^x \log 2$

$g^1(x) = 3^x \log 3$

$f^1(0) = \log 2,$

$g^1(0) = \log 3$

$a \to r, b \to s, c \to q, d \to p$

33. $f^1(x) = \frac{\sqrt{a}}{2\sqrt{x}} - \frac{a^2}{2\sqrt{a}} \times \frac{1}{x\sqrt{x}} \Rightarrow f^1(a) = 0$

34. $f^1(x) = \lim_{x\to 0} \frac{x^p \cos\left(\frac{1}{x}\right) - 0}{x-0} = \lim_{x\to 0} x^{p-1}\cos\left(\frac{1}{x}\right)$. $\therefore p > 1$

35. $f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \ldots \Rightarrow f^1(x) = 1 + x + \frac{x^2}{2!} + \cdots \ldots \Rightarrow f^1(0) = 1.$

WORKSHEET-2

Level – I

1. C 2. A 3. D 4. A 5. A 6. D 7. A

8. B 9. C

1. $x^2 + y^2 = a^2 \Rightarrow 2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-x}{y}$

2. $3x^2 + 4xy + 2y^2 = 0 \Rightarrow 6x + 4x\frac{dy}{dx} + 4y + 4y\frac{dy}{dx} = 0$

$\frac{dy}{dx}(4x+4y) = -(6x+4y) \Rightarrow \frac{dy}{dx}\text{ at } (-1,3) = \frac{-(-6+12)}{-4+12} = \frac{-3}{4}$

3. $y = x^x \Rightarrow \log y = x\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{x}{x} + \log x \Rightarrow \frac{dy}{dx} = y(1+\log x)$

4. $y = at^2, x = 5at^3 \Rightarrow \frac{dy}{dt} = 2at, \frac{dx}{dt} = 15at^2 \Rightarrow \frac{dy}{dx} = \frac{2at}{15at^2} = \frac{2}{15t}$

5. $x = at^2, y = 2at \Rightarrow \frac{dx}{dt} = 2at\ \frac{dy}{dt} = 2a \Rightarrow \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$

6. $\text{x} = a\text{t}^2, \text{y} = 2\text{at} \Rightarrow \frac{dx}{dt} = 2at, \frac{\text{dy}}{\text{dt}} = 2\text{a} \Rightarrow \frac{\text{dy}}{\text{dx}} = \frac{1}{\text{t}} \Rightarrow \frac{\text{d}^2\text{y}}{\text{dx}^2} = \frac{-1}{2at^3}$

7. $f(x) = \frac{a^x}{x^a} \Rightarrow f'(x) = \frac{1}{x^a}a^x\log a + a^x(-ax^{-a-1})$

$f'(a) = \frac{a^a \log a}{a^a} - a^a a \cdot a^{-a-1} = \log a - 1$

8. $u = a^x; v = \sin^{-1}(x) \Rightarrow \frac{du}{dx} = a^x\log a; \frac{dv}{dx} = \frac{1}{\sqrt{1-x^2}} \Rightarrow \frac{du}{dv} = \frac{a^x\log a}{\sqrt{1-x^2}}$

9. $x = \sin^{-1} t; y = \log(1 - t^2)$

$\frac{dx}{dt} = \frac{1}{\sqrt{1-t^2}}; \frac{dy}{dt} = \frac{1}{1-t^2}(-2t) \Rightarrow \frac{dy}{dx} = \frac{-2t}{1-t^2} \times \sqrt{1-t^2}$

$\frac{dy}{dx} = \frac{-2t}{\sqrt{1-t^2}} \Rightarrow \frac{dy}{dx}$ at $t = \frac{1}{2} = \frac{-2\frac{1}{2}}{\sqrt{1-\frac{1}{4}}} = \frac{-2}{\sqrt{3}}$

Level – II

1. C	2. C	3. C	4. B	5. D	6. C	7. 1
8. D	9. C	10. C	11. 1	12. A	13. A	14. A
15. 2	16. C	17. D	18. B	19. B	20. 1	21. D
22. D	23. B	24. A				

1. $3x^2 - 4xy + 1 = 0$

$\Rightarrow 6x - 4x\frac{dy}{dx} - 4y = 0$

$\Rightarrow 4x\frac{dy}{dx} = 6x - 4y \Rightarrow \frac{dy}{dx} = \frac{3x-2y}{2x} \Rightarrow \frac{dy}{dx}(1,1) = \frac{1}{2}$

2. $x = a(t + \sin t)\ y = a(1 - \cos t)$

$\frac{dx}{dt} = a(1 + \cos t)\ \frac{dy}{dt} = a(\sin t)$

$\frac{dy}{dx} = \frac{a\sin t}{a(1+\cos t)} = \frac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}} = \tan\frac{t}{2}$

$\frac{\mathrm{dx}}{\mathrm{dy}} = \cot\frac{\mathrm{t}}{2}, \therefore \mathrm{p} = \frac{\mathrm{t}}{2}$

3. $x = a\left(\cos t + \log\left(\tan\frac{t}{2}\right)\right), y = a\sin t$

$\Rightarrow \frac{dx}{dt} = a\left(-\sin t + \frac{1}{\mathrm{Tan}\frac{t}{2}}\sec^2\frac{t}{2}\cdot\frac{1}{2}\right) = a\left(-\sin t + \frac{1}{\sin t}\right)$

$= a\left(\frac{1-\sin^2 t}{\sin t}\right) = a\frac{\cos^2 t}{\sin t}.$

$\frac{dy}{dt} = a\cos t$

$\frac{dy}{dx} = a\cos t \times \frac{\sin t}{a\cos^2 t} = \tan t$

4. $u = e^{\sin^{-1} x}\ v = \sin^{-1} x$

$\frac{du}{dx} = e^{\sin^{-1} x}\frac{1}{\sqrt{1-x^2}}\ \frac{dv}{dx} = \frac{1}{\sqrt{1-x^2}}$

$\frac{du}{dv} = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}} \times \sqrt{1-x^2} = e^{\sin^{-1} x}$

5. $x^y = \log x$ then

$y\log x = \log(\log x)$

$\Rightarrow y \cdot \frac{1}{x} + \log x \cdot \frac{dy}{dx} = \frac{1}{\log x} \cdot \frac{1}{x}$

$\Rightarrow \frac{dy}{dx}$ at $x = e = \frac{y}{e} + \frac{dy}{dx} = \frac{1}{e} \Rightarrow \frac{dy}{dx} = \frac{1}{e}$

6. $y = 1 + \sin x$

$\Rightarrow \frac{dy}{dx} = \cos x \Rightarrow \frac{d^2y}{dx^2} = -\sin x$

7. $\mathrm{u} = \sin^{-1}\mathrm{x}, \mathrm{v} = \cos^{-1}\sqrt{1-\mathrm{x}^2}$

Let $x = \sin\theta$

$\Rightarrow \mathrm{u} = \theta \Rightarrow \frac{\mathrm{du}}{\mathrm{d}\theta} = 1; \mathrm{v} = \cos^{-1}(\cos\theta) \Rightarrow \mathrm{v} = \theta \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}\theta} = 1$

$\Rightarrow \frac{\mathrm{du}}{\mathrm{dv}} = 1$

8. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\cos^{-1}\mathrm{x} + \sin^{-1}\sqrt{1-\mathrm{x}^2}\right)$ (Put $\mathrm{x} = \cos\theta \Rightarrow \theta = \cos^{-1}\mathrm{x}$)

$= \frac{\mathrm{d}}{\mathrm{dx}}(\theta + \theta) = \frac{\mathrm{d}}{\mathrm{dx}}(2\cos^{-1}\mathrm{x}) = \frac{-2}{\sqrt{1-\mathrm{x}^2}}$

9. $y = x^y \Rightarrow \log y = y\log x$

$\Rightarrow \frac{1}{y}\frac{dy}{dx} = y \cdot \frac{1}{x} + \log x \frac{dy}{dx} \Rightarrow \left(\frac{1}{y} - \log x\right)\frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1 - y\log x)}$

10. Let $\mathrm{y} = (\mathrm{x}^{\mathrm{x}})^{\mathrm{x}} \Rightarrow \log\mathrm{y} = \mathrm{x}\log\mathrm{x}^{\mathrm{x}}$

$\log y = x^2\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = x^2 \cdot \frac{1}{x} + \log x(2x)$

$\frac{dy}{dx} = y(x + 2x\log x) = (x^x)^x x[1 + 2\log x].$

11. $e^y + xy = e^x \Rightarrow e^y\frac{dy}{dx} + x\frac{dy}{dx} + y = e^x$

$\Rightarrow \frac{dy}{dx}(e^y + x) = e^x - y \Rightarrow \frac{dy}{dx} = \frac{e^x - y}{e^y + x} \Rightarrow \frac{dy}{dx}(0,0) = 1$

12. $\frac{\mathrm{d}}{\mathrm{dx}}\left(\cos^{-1}\mathrm{x} + \cos^{-1}\sqrt{1-\mathrm{x}^2}\right)$. Let $\mathrm{x} = \sin\theta \Rightarrow \theta = \sin^{-1}\mathrm{x}$

$\frac{\mathrm{d}}{\mathrm{dx}}(\cos^{-1}(\sin\theta) + \cos^{-1}(\cos\theta)) = \frac{\mathrm{d}}{\mathrm{dx}}(\cos^{-1}(\cos(90 - \theta)) + \theta) = \frac{\mathrm{d}}{\mathrm{dx}}(90 - \theta + \theta) = 0$

13. $x = a\cos^4 t, y = b\sin^4 t \Rightarrow \frac{dx}{dt} = 4a\cos^3 t(-\sin t), \frac{dy}{dt} = 4b\sin^3 t\cos t$

$\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{4\ \mathrm{bsin}^3\,\mathrm{tcos\,t}}{-4\mathrm{acos}^3\,\mathrm{tsin\,t}} = \frac{-b}{a}\tan^2 t \Rightarrow \frac{\mathrm{dy}}{\mathrm{dx}}\mathrm{att} = \frac{3\pi}{4} = \frac{-\mathrm{b}}{\mathrm{a}}\mathrm{Tan}^2\left(\pi - \frac{\pi}{4}\right) = \frac{-\mathrm{b}}{\mathrm{a}}$

14. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$\Rightarrow 2ax + 2by\frac{dy}{dx} + 2hy + 2hx\frac{dy}{dx} + 2g + 2f\frac{dy}{dx} = 0$

$\frac{dy}{dx}(hx + by + f) = -(ax + hy + g) \Rightarrow \frac{dy}{dx} = \frac{-(ax + hy + g)}{(hx + by + f)}$

15. Let $u = \text{Tan}^{-1}\frac{\sqrt{1+x^2}-1}{x}$; $v = \text{Tan}^{-1}x$. Put $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$.

$$u = \text{Tan}^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right) = \text{Tan}^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right) = \text{Tan}^{-1}\left(\tan\frac{\theta}{2}\right) = \frac{\theta}{2}$$

$\Rightarrow u = \frac{\theta}{2} \Rightarrow u = \frac{1}{2}\text{Tan}^{-1}x$. Then, $\frac{du}{d\theta} = \frac{1}{2(1+x^2)}$.

$v = \text{Tan}^{-1}(\tan\theta) = \tan^{-1}x$. Then, $\frac{dv}{d\theta} = \frac{1}{1+x^2}$.

$$\Rightarrow \frac{du}{dv} = \frac{1}{2(1+x^2)} \times (1+x^2) = \frac{1}{2}$$

16. $x = \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$ $y = \sin^{-1}\left(\frac{t}{\sqrt{1+t^2}}\right)$

Let $t = \tan\theta \Rightarrow x = \cos^{-1}\left(\frac{1}{\sec\theta}\right)$ &$y = \sin^{-1}\left(\frac{\text{Tan}\,\theta}{\sec\theta}\right)$

$x = \theta$&$y = \sin^{-1}(\sin\theta) = \theta$. Then, $\frac{dx}{d\theta} = \frac{dy}{d\theta} = 1 \Rightarrow \frac{dy}{dx} = 1$

17. $x = \cos^3\theta, y = \sin^3\theta$

$$\frac{dx}{d\theta} = 3\cos^2\theta(-\sin\theta),\ \frac{dy}{d\theta} = 3\sin^2\theta\cos\theta \Rightarrow \frac{dy}{dx} = \frac{3\sin^2\theta\cos\theta}{-3\cos^2\theta\sin\theta} = -\text{Tan}\,\theta$$

$$\left(\frac{dy}{dx}\right)^2 = \text{Tan}^2\theta \Rightarrow \sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{1+\text{Tan}^2\theta} = |\sec\theta|$$

18. $x^y = e^{x-y} \Rightarrow \log x^y = \log e^{x-y} \Rightarrow y\log x = (x-y)\log_e^e \Rightarrow y\frac{1}{x} + \log x \cdot \frac{dy}{dx} = 1 - \frac{dy}{dx}$

$$(1+\log x)\frac{dy}{dx} = 1 - \frac{y}{x} = \frac{x-y}{x}$$

$$\frac{dy}{dx} = \frac{x-y}{x(1+\log x)} = \frac{y\log x}{x(1+\log x)} = \frac{y\log x}{y(1+\log x)^2}\ (y\log x + y = x;\ y(1+\log y) = x)$$

$$= \frac{\log x}{(1+\log x)^2}$$

19. $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots \ldots \ldots \propto}}}$

$$y^2 = \sin x + \sqrt{\sin x + \sqrt{\sin x} + \cdots \ldots} \Rightarrow y^2 = \sin x + y$$

$$\Rightarrow 2y\frac{dy}{dx} = \cos x + \frac{dy}{dx} \Rightarrow \frac{dy}{dx}(2y-1) = \cos x \Rightarrow \frac{dy}{dx} = \frac{\cos x}{2y-1}$$

20. $x + y = \sin(x+y)$

$$\Rightarrow 1 + \frac{dy}{dx} = \cos(x+y)\left(1+\frac{dy}{dx}\right) \Rightarrow 1 + \frac{dy}{dx} = \cos(x+y) + \cos(x+y)\frac{dy}{dx}$$

$$\Rightarrow (1-\cos(x+y))\frac{dy}{dx} = \cos(x+y) - 1 \Rightarrow \frac{dy}{dx} = \frac{\cos(x+y)-1}{-[\cos(x+y)-1]} = -1$$

21. If $x = \cos\theta, y = \sin 5\theta \Rightarrow \frac{dx}{d\theta} = -\sin\theta\ \frac{dy}{d\theta} = 5\cos 5\theta$

$$\frac{dy}{dx} = \frac{5\cos 5\theta}{-\sin\theta} = \frac{5\sqrt{1-\sin^2 5\theta}}{-\sqrt{1-\cos^2\theta}} \Rightarrow \frac{dy}{dx} = \frac{5\sqrt{1-y^2}}{-\sqrt{1-x^2}} \Rightarrow (1-x^2)\left(\frac{dy}{dx}\right)^2 = 25(1-y^2)$$

$\Rightarrow (1-x^2)2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2}-\left(\frac{dy}{dx}\right)^2 2x=-50y\frac{dy}{dx}\Rightarrow(1-x^2)\frac{d^2y}{dx^2}-x\frac{dy}{dx}=-25y$

22. $\frac{x^n}{a^n}+\frac{y^n}{b^n}=6\Rightarrow\Rightarrow\frac{1}{a^n}n\cdot x^{n-1}+\frac{1}{b^n}ny^{n-1}\frac{dy}{dx}=0\Rightarrow\frac{dy}{dx}=-\frac{x^{n-1}}{a^n}\times\frac{b^n}{y^{n-1}}$

Then, $\frac{dy}{dx}(a,b)=\frac{-a^{n-1}}{a^n}\times\frac{b^n}{b^{n-1}}=\frac{-b}{a}$

23. $y=\sqrt{\text{Tan}\,x+\sqrt{\tan x+}}\,\infty$

$\Rightarrow y^2=\text{Tan}\,x+y\Rightarrow 2y\frac{dy}{dx}=\sec^2 x+\frac{dy}{dx}\Rightarrow\frac{dy}{dx}=\frac{\sec^2 x}{2y-1}$

24. $\text{x}^{\text{y}}=\text{y}^{\text{x}}\Rightarrow\log x^y=\log y^{\text{x}}\Rightarrow\text{ylog x}=\text{xlog y}$

$\Rightarrow y\frac{1}{x}+\log x\frac{dy}{dx}=x\cdot\frac{1}{y}\frac{dy}{dx}+\log y\Rightarrow\left(\frac{x}{y}-\log x\right)\frac{dy}{dx}=-\log y+\frac{y}{x}$

$\Rightarrow\frac{(x-y\log x)}{y}\frac{dy}{dx}=\frac{-x\log y+y}{x}\Rightarrow x(x-y\log x)\frac{dy}{dx}=y(-x\log y+y)$

Level – III

25. A,B	26. A	27. A	28. B	29.a-q; b-p; c-s; d-t
30. 3	31. B,C	32. 1	33. C,D	34. 1

25

1) $y=(\log x)^x\Rightarrow\log y=x\log(\log x)$

$\Rightarrow\frac{1}{y}\frac{dy}{dx}=x\frac{1}{\log x}\frac{1}{x}+\log(\log x)\Rightarrow\frac{dy}{dx}=y\frac{[1+\log x\cdot\log(\log x)]}{\log x}$

$=(\log x)^{x-1}(1+\log x\cdot\log(\log x))$

2) $y=x^x\Rightarrow\log y=x\log x\Rightarrow\frac{1}{y}\frac{dy}{dx}=x\cdot\frac{1}{x}+\log x$

$\frac{dy}{dx}=x^x(1+\log x)=x^x(\log ex)$

3. $y=x^y\Rightarrow\log y=y\log x\Rightarrow\frac{1}{y}\frac{dy}{dx}=y\cdot\frac{1}{x}+\log x\frac{dy}{dx}\Rightarrow\frac{dy}{dx}=\frac{y^2}{x(1-y\log y)}$

26. Let $y=t^{10}+1$ and $x=t^8+1$ then $\frac{d^2y}{dx^2}$ is

$\frac{dy}{dt}=10t^9,\ \frac{dx}{dt}=8t^7\Rightarrow\frac{dy}{dx}=\frac{10t^9}{8t^7}\Rightarrow\frac{dy}{dx}=\frac{10t^9}{8t^7}$

$\frac{dy}{dx}=\frac{5}{4}t^2\Rightarrow\frac{dy}{dx}=\frac{5}{4}(x-1)^{1/4}\Rightarrow\frac{d^2y}{dx^2}=\frac{5}{16t^6}$

27. $y=\sin x+e^x\Rightarrow y^1=\cos x+e^x\Rightarrow y^{11}=-\sin x+e^x\Rightarrow\frac{d^2x}{dy^2}=(-\sin x+e^x)^{-1}$

28. $y=x+e^x\Rightarrow 1=\frac{dx}{dy}+e^x\frac{dx}{dy}\Rightarrow 1=\frac{dx}{dy}(1+e^x)\Rightarrow\frac{dx}{dy}=\frac{1}{1+e^x}$

$\frac{dx}{dy}=(1+e^x)^{-1}\Rightarrow\frac{d^2x}{dy^2}=-(1+e^x)^{-2}\left(e^x\frac{dx}{dy}\right)\Rightarrow\frac{d^2x}{dy^2}=\frac{-e^x}{(1+e^x)^3}$

29. a) $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}\Rightarrow\frac{1}{\sqrt{1-x^2}}+\frac{1}{\sqrt{1-y^2}}\frac{dy}{dx}=0\Rightarrow\frac{dy}{dx}=\frac{-\sqrt{1-y^2}}{\sqrt{1-x^2}}$

b) $\text{Tan}^{-1}\text{x}+\text{Tan}^{-1}\text{y}=\frac{\pi}{2}\Rightarrow\frac{1}{1+\text{x}^2}+\frac{1}{1+\text{y}^2}\frac{\text{dy}}{\text{dx}}=0\Rightarrow\frac{\text{dy}}{\text{dx}}=-\frac{(1+\text{y}^2)}{1+\text{x}^2}$

c) $\log(x^m y^n) = \log(x+y)^{m+n} \Rightarrow m\log x + n\log y = (m+n)\log(x+y) \Rightarrow \frac{dy}{dx} = \frac{y}{x}$

d) $x = at^2\ y = 2at$

$\frac{dx}{dt} = 2at\ \frac{dy}{dt} = 2a \Rightarrow \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t} \Rightarrow \frac{d^2y}{dx^2} = \frac{-1}{t^2}$

30. $\sqrt{x} + \sqrt{y} = 4$

$\Rightarrow \frac{dx}{dy} = \frac{\sqrt{y}-4}{\sqrt{y}}$

$\Rightarrow \frac{1}{2\sqrt{x}}\frac{dx}{dy} + \frac{1}{2\sqrt{y}} = 0$

$\frac{dx}{dy} = \frac{-\sqrt{x}}{\sqrt{y}} \Rightarrow \frac{dx}{dy}$ at $y = 1 = -3$

31. $f(x) = ax^2 + bx + c \Rightarrow f^1(x) = 2ax + b \Rightarrow f^1(5) = 10a + b$

32. $\mathbf{x} = \mathrm{Tan}\,\theta$

$\mathrm{u} = \mathrm{Tan}^{-1}\left(\frac{2\mathrm{Tan}\,\theta}{1-\mathrm{Tan}^2\,\theta}\right) = 2\theta \Rightarrow \frac{\mathrm{du}}{\mathrm{d}\theta} = 2.$

$v = \sin^{-1}(\sin 2\theta) \Rightarrow v = 2\theta \Rightarrow \frac{dv}{d\theta} = 2.$ Then $\frac{du}{dv} = 1.$

33. $\mathrm{f}(\mathrm{x}) = \sin^{-1}(\sin \mathrm{x}) \Rightarrow \mathrm{f}(\mathrm{x}) = \begin{cases} \pi - \mathrm{x}, & \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \\ \mathrm{x} - 2\pi, & \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \end{cases} \Rightarrow \mathrm{f}^1(\mathrm{x}) = \begin{cases} -1, & \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) \\ 1, & \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \end{cases}$

34. $f(x) = \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x$

$f(x) = \frac{2\sin x \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x}{2\sin x} \Rightarrow f(x) = \frac{\sin 32x}{2^5 \sin x}$

$f^1(x) = \frac{1}{32}\left[\frac{32\cos 32x \cdot \sin x - \cos x \cdot \sin 32x}{\sin^2 x}\right]$

$\Rightarrow f^1\left(\frac{\pi}{4}\right) = \frac{32 \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}(0)}{32 \times \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{2}.$

3. ERROR'S AND APPROXIMITION AND RATE MEASURE

WORKSHEET-1

Level – I

1. B	2. A	3. A	4. B	5.A	6. A	7. A
8. A	9. D	10. C				

Level – II

1. 3	2. 1	3. 1	4. 2	5. 1	6. 1	7. 2
8. 3	9. 3	10. 3	11. 2	12. 3	13. 1	14. 2
15. 5	16. 4	17. 2	18. 3	19. 54	20. 3	21. 3
22. 96	23. 2,4	24. 4	25. 2	26. 1	27. a-r; b-p; c-s; d-q	
28. 0						

1. Radius of sphere, $r = 3$ cm, $\delta r = -0.02$

   volume, $v = \frac{4}{3}\pi r^3$

   $\mathrm{dv} = \frac{4}{3}\pi \times 3\mathrm{r}^2\delta\mathrm{r} = -4\pi \times 3 \times 3 \times 0.02 = -0.72\pi\mathrm{cm}^3$.

2. Surface area of sphere, $A = 4\pi r^2 = A\% = 2 \times r\%$

   $\mathrm{K} = 2\mathrm{r}\% \Rightarrow \mathrm{r}\% = \frac{\mathrm{K}}{2}$

   Volume, $V = \frac{4}{3}\pi r^3$

   $\mathrm{V}\% = 3\mathrm{r}\% = \frac{3\,\mathrm{K}}{2}$.

3. Percentage error is radius, $r\% = 0.1$

   Volume of sphere, $V = \frac{4}{3}\pi r^3$

   $\mathrm{V}\% = 3\mathrm{r}\% = 3 \times 0.1 = 0.3\%$.

4. Mass of the body, $m = 100$ kg

   $s = 2t^2 + 3t + 1$

   Velocity, $V = \frac{ds}{dt} = 4t + 3$; Velocity after 5sec = 23 m/sec

   $\mathrm{K.E} = \frac{1}{2}\mathrm{mv}^2 = \frac{1}{2} \times 100 \times 23 \times 23 = 26450$.

5. Equation of ellipse is $16x^2 + 9y^2 = 400$

   diff. w.r.t t

   $32x\frac{dx}{dt} + 18y\frac{dy}{dt} = 0 \Rightarrow 18y\frac{dy}{dt} = -32x\frac{dx}{dt}$

   $18\mathrm{y} = 32\mathrm{x}\ \left(\because \frac{\mathrm{dy}}{\mathrm{dt}} = -\frac{\mathrm{dx}}{\mathrm{dt}}\right)$

   $y = \frac{32}{18}x = \frac{16}{9}x$ $\therefore$ Point $= \left(3, \frac{16}{3}\right)$.

6. Let $f(x)=\frac{1}{\sqrt{x}}, x=25, \delta x=0.25$

   Approximate value of $\frac{1}{\sqrt{25.25}}=f(x)+f^1(x)\delta x$

   $=\frac{1}{\sqrt{x}}-\frac{1}{2}x^{-3/2}\delta x=\frac{1}{\sqrt{25}}-\frac{1}{2}\cdot\frac{1}{(25)^{3/2}}(0.25)=\frac{1}{5}-\frac{1}{2}\cdot\frac{1}{125}\times 0.25=0.2-0.001=0.199.$

7. Let $f(x)=\sqrt[4]{x}, x=81, \sqrt{x}=-1$

   Approximate value of $\sqrt[4]{80}$ is $=f(x)+f^1(x)\delta x$

   $=x^{1/4}+\frac{1}{4}x^{-3/4}\cdot\delta x\Rightarrow(81)^{1/4}+1/4\cdot\frac{1}{(81)^{3/4}}(-1)$

   $=3-\frac{1}{4}\cdot\frac{1}{27}\Rightarrow 3-0.009259=2.9907\Rightarrow[2.9907]=2.$

8. Period of simple pendulum is

   $T=4\pi\sqrt{\frac{l}{g}}\Rightarrow T\%=\frac{1}{2}l\%\Rightarrow a=\frac{1}{2}l\%\ \therefore 1\%=2a.$

9. $PV^{1/4}=C\Rightarrow P\%+\frac{1}{4}\ V\%=0\Rightarrow P\%=-\frac{1}{4}\ V\%\Rightarrow-\frac{1}{4}\left(-\frac{1}{2}\right)=\frac{1}{8}.$

10. Equation of curve is $y=x^2+2x\Rightarrow\frac{dy}{dt}=(2x+2)\frac{dx}{dt}$

    $\Rightarrow 2x+2=1\Rightarrow x=-1/2\Rightarrow y=\frac{1}{4}-1=-3/4\ \therefore(x,y)=(-1/2,-3/4).$

11. Let $f(x)=\sin x, x=45°, \delta x=1^0$

    Approximate value of $\sin 46°$

    $=f(x)+f^1(x)\delta x=\sin x+\cos x\cdot\delta x=\sin 45+\cos 45\cdot 1°=\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\times 0.01745.$

12. Volume of cone is $V=\frac{1}{3}\pi r^2h\Rightarrow V\%=2\times r\%+h\%=2\times 1+1=3.$

13. Area of triangle, $s=\frac{1}{2}bc\sin A$

    $\Rightarrow\delta S=\frac{1}{2}bc\cos A\delta A\Rightarrow\frac{\delta S}{S}=\cot A\delta A\Rightarrow\delta A=15^1=\frac{15}{60}\times\frac{\pi}{180}$

    $\Rightarrow\frac{\delta S}{s}\times 100=\cot 63°\times\frac{15}{60}\times\frac{\pi}{180}\times 100=\frac{5\pi}{36}\cot 63°.$

14. $a^2=b^2+c^2-2bc\cos A\Rightarrow 2a\delta a=2bc\sin A\delta A$

    $\Rightarrow\delta a=\frac{bc\sin A\delta A}{a}=\frac{2\Delta\delta A}{a}.\ \left(\because\Delta=\frac{1}{2}bc\sin A\right)$

15. Area of circular plate, $A=\pi r^2\Rightarrow\delta A=2\pi r\delta r$

    $\Rightarrow\frac{\pi}{10}=2\pi r\times 0.01\Rightarrow r=\frac{1}{10\times 2\times 0.01}=5$ cm.

16. $x^2+y^2=100\Rightarrow 6^2+y^2=100\Rightarrow y=8m$

    $\Rightarrow 2x\frac{dx}{dt}+2y\frac{dy}{dt}=0\Rightarrow 6\times 2+8\frac{dy}{dt}=0\Rightarrow\frac{dy}{dt}=\frac{-12}{8}=\frac{-3}{2}$ m/min $\Rightarrow A=\frac{1}{2}xy$

    $\Rightarrow\frac{dA}{dt}=\frac{1}{2}\left(x\frac{dy}{dt}+y\frac{dx}{dt}\right)\Rightarrow\frac{1}{2}\left[6\left(-\frac{3}{2}\right)+8(2)\right]\Rightarrow\frac{1}{2}(-9+16)=\frac{7}{2}\text{sq}\cdot\text{m/min}$

17. Volume of ball, $V=\frac{4}{3}\pi r^3\Rightarrow 288\pi=\frac{4}{3}\pi r^3\Rightarrow r^3=\frac{288\times 3}{4}=72\times 3=216$

    $\therefore r=6$ cm

    $\Rightarrow\frac{dV}{dt}=\frac{4}{3}\pi\times 3r^2\frac{dr}{dt}\Rightarrow 2\pi=4\pi\times 6\times 6\times\frac{dr}{dt}\Rightarrow\frac{dr}{dt}=\frac{1}{72}$ cm/sec.

18. Let radius of sphere be r and Height of cone be h

$\therefore h = r$

Volume of ballon $= \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 \Rightarrow V = \pi r^3 \Rightarrow \frac{dV}{dt} = 3\pi r^2 \frac{dr}{dt} = 3\pi \times (2)^2 \frac{dr}{dt} = 12\pi \frac{dr}{dt}$.

19. $y = x^3 \Rightarrow \frac{dy}{dx} = 3x^2 \Rightarrow m = 3x^2 \Rightarrow \frac{dm}{dt} = 6x\frac{dx}{dt} = 6 \times 3 \times 3 = 54$.

20. $\theta \alpha t^2$

$\Rightarrow \theta = kt^2 \Rightarrow 2\pi = k \times 8^2 \Rightarrow k = \frac{2\pi}{8\times 8} \Rightarrow k = \frac{\pi}{32} \Rightarrow \frac{d\theta}{dt} = k \cdot 2t = \frac{\pi}{32} \times 2 \times 8 = \frac{\pi}{2}$ rad/sec.

21. $\frac{h}{r} = \frac{10}{5} \Rightarrow \frac{h}{2} = r$

Volume, $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times \frac{h^3}{4}$

$\therefore V = \frac{\pi}{12} h^3$.

$\Rightarrow \frac{dV}{dt} = \frac{\pi}{12} \times 3 \times h^2 \frac{dh}{dt} \Rightarrow \frac{3}{2} = \frac{\pi}{12} \times 3 \times 6 \times 6 \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{3}{2\times 4\pi} = \frac{3}{8\pi}$ mt/min.

22. $s = 160t - 16t^2 \Rightarrow v = \frac{ds}{dt} = 160 - 32t$

At $s = 256 \Rightarrow 160t - 16t^2 = 256$

$\Rightarrow 16t^2 - 160t + 256 = 0 \Rightarrow t^2 - 10t + 16 = 0$

$\therefore (t-2)(t-8) = 0 \ \therefore t = 2,8.$

$\Rightarrow V(2) = 160 - 322 = 96 \Rightarrow V(8) = 160 - 32(8) = -96$

Velocity on the way up = 96 m/sec.

23. $s = \frac{t^3}{3} - 3t^2 + 8t + 5$& $V = \frac{ds}{dt} \Rightarrow V = t^2 - 6t + 8$

Direction of particle changes when $V = 0$

$\Rightarrow t^2 - 6t + 8 = 0 \Rightarrow (t-2)(t-4) = 0 \ \therefore t = 2,4.$

24. $\frac{dr}{dt} = 1$ cm/sec $\Rightarrow$ Area of circle, $A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi \times 5 \times 1 = 10\pi$.

25. $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi \times 3 \times 1 = 6\pi$.

26. Perimeter, $P = 2\pi r \Rightarrow \frac{dP}{dt} = 2\pi \frac{dr}{dt} = 2\pi \times 1 = 2\pi$.

27. a) $x^2 + y^2 = 2 \Rightarrow 2x\frac{dy}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow y\frac{dy}{dt} = -x\frac{dx}{dt} = 0 \Rightarrow y = -x \ \left(\because \frac{dy}{dt} = \frac{dx}{dt}\right)$

By verification, point $= (1, -1)$.

b) $x^2 = 8y \Rightarrow 2x\frac{dx}{dt} = 8\frac{dy}{dt} \Rightarrow x = 4 \& y = 2.$ $\therefore$ point $= (4,2)$.

c) $x^2 - y^2 = 0 \Rightarrow 2x\frac{dx}{dt} - 2y\frac{dy}{dt} = 0 \Rightarrow x\frac{dx}{dt} = y\frac{dy}{dt} \Rightarrow x = y.$ $\therefore$ point $= (4,4)$.

d) $y^2 = 8x \Rightarrow 2t\frac{dy}{dt} = 8\frac{dx}{dt} \Rightarrow y = 4 \& x = 2.$

$\therefore$ point $= (2,4)$.

28. $s = t^3 - 3t^2 + 3t + 12 \Rightarrow v = \frac{ds}{dt} = 3t^2 - 6t + 3$

$a = \frac{dv}{dt} = 6t - 6; a = 0 \Rightarrow 6t - 6 = 0 \Rightarrow t - 1 = 0 \Rightarrow t = 1 \ \therefore v(1) = 3 - 6 + 3 = 0$

Level – III

29. A,C 30. 1 31. D 32. D 33. C

34. a-s; b-p; c-q; d-r 35. 1,2,3

29. $y = \frac{x^3}{3} - \frac{3}{2}x^2 + 5x + 4 \Rightarrow \frac{dy}{dt} = (x^2 - 3x + 5)\frac{dx}{dt} \Rightarrow 3\frac{dx}{dt} = (x^2 - 3x + 5)\frac{dx}{dt}$

$\Rightarrow x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0 \ \therefore x = 1,2.$

30. $s = t^3 - 2t^2 - 3t + 5 \Rightarrow v = \frac{ds}{dt} = st^2 - 4t - 3 \Rightarrow \therefore v(2) = 3(2)^2 - 4(2) - 3 = 1.$

31. Let $f(x) = \sin x, x = 60°, \delta x = 6^1$

Approximation value of $\sin 60° 6^1$

$= f(x) + f^1(x)\delta x = \sin x + \cos x \cdot \delta x = \sin 60° + \cos 60° \cdot \frac{6}{60} \times 0.0175$

$= \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{10} \times 0.0175 = 0.86603 + 0.00087 = 0.8669.$

32. Let $f(x) = \tan x, x = 60°, \delta x = 6^1$

Approxomate value of $\tan 60° 6^1$

$= \tan x + \sec^2 x\delta x = \tan 60 + \sec^2 60 \times \frac{6}{60} \times 0.0175 = \sqrt{3} + 2 \times \frac{1}{10} \times 0.0175$

$= 1.732 + 0.0035 = 1.7355.$

33. Let $f(x) = \sin x, x = 60°, \delta x = 1^1$

Approximate value of $\sin 60° 1^1$

$= \sin 60° + \cos 60° \times \frac{1}{60} \times 0.01745 = \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{60} \times 0.01745 = 0.86603 + 0.00015.$

$\therefore 0.86618$

34. a) Let $f(x) = \log x, x = 4, \delta x = 0.01$

Approximate value of $\log 4.01 = \log x + \frac{1}{x}\delta x = \log 4 + \frac{1}{4} \times 0.01 = 1.3868 + 0.0025 = 1.3893.$

b) Let $f(x) = \tan x, x = 45°, \delta x = -1^0$

Approximate value of $\tan 44° = \tan 45 - \sec^2 45 \times 0.01745$

$= 1 - 2 \times 0.01745 = 1 - 0.0349 = 0.9651.$

c) Let $f(x) = \frac{1}{\sqrt{x}}, x = 25, \delta x = 0.25$

Approximate value of $\frac{1}{\sqrt{25.25}} = \frac{1}{\sqrt{x}} - \frac{1}{2}x^{-3/2}\delta x$

$= \frac{1}{25} - \frac{1}{2}\frac{1}{(25)^{3/2}}(0.25) \Rightarrow \frac{1}{5} - \frac{1}{2} \cdot \frac{1}{125} \times 0.25 = 0.2 - 0.001 = 0.199.$

d) Let $f(x) = \sqrt[6]{x}, x = 729, \delta x = -1$

Approximate value of $\sqrt[6]{728}$

$= x^{1/6} + \frac{1}{6}x^{-5/6}.\delta x = (729)^{1/6} + \frac{1}{6}\frac{1}{(729)^{5/6}}(-1) = 3 - \frac{1}{6} \times \frac{1}{243} \Rightarrow 3 - 0.0007 = 2.9993.$

35. $s = at^2 + bt + e \Rightarrow s(1) = a + b + c = 16$

$V = \frac{ds}{dt} = 2at + b \Rightarrow V(2) = 4a + b = 24 - \cdots - (2)$

$a = \frac{dV}{dt} = 2a \Rightarrow 2a = 8 \Rightarrow a = 4 \Rightarrow 16 + b = 24 \Rightarrow b = 8 \Rightarrow 4 + 8 + c = 16 \Rightarrow c = 4$

WORKSHEET-02

Level – I

1. C	2. B	3. C	4. A	5. A	6. B	7. B
8. C	9. D	10. A	11. B			

Level – II

1. B	2. D	3. B	4. D	5. D	6. B	7. 2
8. B	9. A	10. B	11. A	12. 3	13. C	14. D
15. B	16. 9	17. A	18. C	19. D	20. 1	21. B
22. A	23. 0					

1. Equation of curve is $y = 16 - x^2 \Rightarrow \frac{dy}{dx} = -2x$.

   Slope of tangent at (0,16) is $m = 0$

   Equations tangent is $y = 16$.

2. Equation of curve is $\left(\frac{x}{a}\right)^4 + \left(\frac{y}{b}\right)^4 = 2$

   On differentiating both the sides, w.r.t.x,

   $\Rightarrow \frac{4x^3}{a^4} + \frac{4y^3}{b^4}\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{4x^3}{a^4} \times \frac{b^4}{4y^3}$

   slope of tangent at (a, b) is $\Rightarrow m = \frac{-4a^3}{a^4} \cdot \frac{b^4}{4b^3} = -\frac{b}{a}$

   slope of normal $= \frac{-1}{m} = \frac{a}{b}$

   Equation of normal is $y - b = \frac{a}{b}(x - a) \Rightarrow by - b^2 = ax - a^2 \ \therefore ax - by = a^2 - b^2$.

3. Equation of curves are $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{l^2} + \frac{y^2}{m^2} = 1$

   $\Rightarrow a^2x^2 + b^2y^2 = 1$ and $a_1^2x^2 + b_1^2y^2 = 1$

   cut orthogonally, then $\frac{1}{a} - \frac{1}{a_1} = \frac{1}{b} - \frac{1}{b_1} \Rightarrow a^2 - l^2 = b^2 - m^2 \Rightarrow a^2 - b^2 = l^2 - m^2$.

4. Equation of curve is $f(x) = x^3 - xy^2 - 4x^2 - xy + 5x + 3y + 1 = 0$

   $\frac{dy}{dx} = -\frac{\partial f}{\partial x} \Big/ \frac{\partial f}{\partial y} = \frac{-[3x^2 - y^2 - 8x - y + 5]}{-2xy - x + 3}$ slope of tangent at $(1, -1)$ is

   $m = \left(\frac{3-1-8+1+5}{2-1+3}\right)$. Equation of tangent is $y = y_1 \Rightarrow y = -1 \Rightarrow y + 1 = 0$.

5. Equation of curve is $xy = a^2 \Rightarrow y = \frac{a^2}{x} \Rightarrow \frac{dy}{dx} = -\frac{a^2}{x^2}$

   slope of tangent at (x, y) is $m = -\frac{a^2}{x^2}$

Length of sub normal $= |y \times m| = \left|y \cdot \left(-\frac{a^2}{x^2}\right)\right| = y \cdot \frac{y^2}{a^2} = \frac{y^3}{a^2}$.

6. $f(x) = (x-a)(x-b) \Rightarrow f(x) = x^2 - (a+b)x + ab \Rightarrow f^1(x) = 2x - (a+b)$

   By Rolle's theorem, $f^1(x) = 0 \Rightarrow 2x - (a+b) = 0 \Rightarrow x = \frac{a+b}{2}$.

7. $f(x) = x(x+3)(x-2) = (x^2+3x)(x-2) = x^3 + x^2 - 6x$.

   $f^1(x) = 3x^2 + 2x - 6 = 10$ [$\because f^1(x) = 10$]

   $\Rightarrow 3x^2 + 2x - 6 = 10 \Rightarrow 3x^2 + 2x - 16 = 0$

   $\Rightarrow 3x^2 - 6x + 8x - 16 = 0 \Rightarrow 3x(x-2) + 8(x-2) = 0$

   $\Rightarrow (x-2)(3x+8) = 0 \ \therefore x = 2$ or $-\frac{8}{3}$.

8. Equation of curve is $x^3 + y^3 = 6xy \Rightarrow f(x,y) = x^3 + y^3 - 6xy$

   $\Rightarrow \frac{dy}{dx} = -\frac{\partial f}{\partial x} / \frac{\partial f}{\partial y} = -\frac{(3x^2 - 6y)}{3y^2 - 6x}$

   Gradient of the curve at (3,3) is $m = -\frac{(27-18)}{27-18} = -1$.

9. Equation of curves are

   $y = x^3 + x + 1;\ 2y = x^3 + 5x \Rightarrow 2x^3 + 2x + 2 = x^3 + 5x \Rightarrow x^3 - 3x + 2 = 0 \ \therefore x = 1;\ y = 3$.

   The curve touch each other at (1,3).

10. $x = 2\sin t;\ y = 2\cos t$

    On differentiating with respect x,

    $\Rightarrow \frac{dx}{dt} = 2\cos t \ \frac{dy}{dt} = -2\sin t$

    $\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt} = \frac{-2\sin t}{2\cos t} = -\tan t$

    Slope of tangent at $t = \frac{\pi}{2}$ is $m = \infty$

    slope of normal $= 0$

    Equation of normal $y - 2\cos\frac{\pi}{2} = 0 \Rightarrow y = 0$.

11. $f(x) = 1 - \sqrt[3]{x^4} = 1 - x^{\frac{4}{3}} \Rightarrow f^1(x) = -\frac{4}{3}x^{1/3}$

    $\Rightarrow -\frac{4}{3}x^{1/3} = 0 [\because f^1(x) = 0] \Rightarrow x = 0$.

12. Equation of curve is $y = x + \frac{4}{x^2} \Rightarrow \frac{dy}{dx} = 1 + 4\left(\frac{-2}{x^3}\right) = 1 - \frac{8}{x^3}$

    slope of tangent $= \frac{x^3 - 8}{x^3}$

    equation of tangent is $y = k$. it is parallel to x-axis

    slope $= 0 \Rightarrow x^3 - 8 = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$

    $\Rightarrow y = 2 + \frac{4}{4} = 3 \ \therefore k = 3$.

13. Let $f(x) = \frac{4ax^4}{4} + \frac{3bx^3}{3} + \frac{2cx^2}{2} + dx$

    $f(x) = ax^4 + bx^3 + cx^2 + dx \Rightarrow f^1(x) = 4ax^3 + 3bx^2 + 2cx + d = 0 \ \therefore f(0) = 0$

    $f(3) = 81a + 27b + 9c + 3d = 3(27a + 9b + 3c + d) = 0 \ \therefore f(0) = 0 = f(3)$

$\exists c \in (0,3)$ such that $f^1(c) = 0$.

14. Equation of curve is $y^2x^2 = a^2(x^2 - a^2) \Rightarrow y^2 = a^2 - \frac{a^4}{x^2} \Rightarrow 2yy^1 = \frac{2a^4}{x^3}$

$\Rightarrow y^1 = \frac{a^4}{x^3y} \Rightarrow$ slope of tangent, $m = \frac{a^4}{x^3y}$

length of subnormal $= |ym| = \left|y\frac{a^4}{x^3y}\right| = \frac{a^4}{x^3}\alpha\frac{1}{x^3}$.

15. Equation of curve is

$x = e^{\sin y} \Rightarrow \log x = \sin y \log e \Rightarrow \log x = \sin y \Rightarrow \frac{1}{x} = \cos y\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{x\cos y}$

slope of tangent at (1,0) is $m = 1$

slope of normal $= -1$

Equation normal is $y - 0 = -1(x - 1)$

$\therefore x + y = 1.$

Area of triangle formed by the normal with co-ordinate axes $= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$ sq.units.

16. $y = ax^2 + bx + c \Rightarrow \frac{dy}{dx} = 2ax + b$

Where $x = 1, y = 0 \Rightarrow a + b + c = 0$

$y = 3x - 3$

$y = x + 1$

$\Rightarrow \frac{dy}{dx} = 3$ at $x = 1$

$\frac{dy}{dx} = 1$ at $x = 3$

$\Rightarrow 2a + b = 3 \Rightarrow 6a + b = 1 \Rightarrow 4a = -2 \Rightarrow a = -\frac{1}{2};\ b = 4$

$\Rightarrow -\frac{1}{2} + 4 + c = 0$

$\therefore c = -\frac{7}{2}$

$2a - b - 4c = 2\left(-\frac{1}{2}\right) - 4 - 4\left(-\frac{7}{2}\right) \Rightarrow -1 - 4 + 14 = 9.$

17. Equation of curves are

$\Rightarrow x = a\left[\cos t + \log\left(\tan\frac{t}{2}\right)\right], y = a\sin t \Rightarrow \frac{dx}{dt} = a\left[-\sin t + \frac{1}{\tan t/2}\sec^2 t/2 \cdot \frac{1}{2}\right]$

$= a\left[-\sin t + \frac{\cos t/2}{\sin t/2} \cdot \frac{1}{\cos^2 t/2} \cdot \frac{1}{2}\right] = a\left[-\sin t + \frac{1}{\sin t}\right] \Rightarrow a\left[\frac{1 - \sin^2 t}{\sin t}\right] = a\frac{\cos^2 t}{\sin t}$

$\therefore \frac{dy}{dt} = a\cos t$

$\Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a\cos t}{a\cos^2 t} \cdot \sin t = \tan t$

slope of tangent at $t=\frac{\pi}{3}$ is $\Rightarrow m=\tan\frac{\pi}{3}=\sqrt{3}$

length of subtangent $=\left|\frac{y_1}{m}\right|=\left|\frac{a\sin\pi/3}{\sqrt{3}}\right|=\frac{a\sqrt{3}}{\sqrt{3}\times 2}=\frac{a}{2}$.

18. $f(0)=2, g(0)=0, f(1)=6, g(1)=2$

From cauchys' mean value theorem

$\Rightarrow \frac{f(b)-f(a)}{g(b)-g(a)}=\frac{f^1(c)}{g^1(c)}\Rightarrow \frac{f(1)-f(0)}{g(1)-g(0)}=\frac{f^1(c)}{g^1(c)}\Rightarrow \frac{6-2}{2-0}=\frac{f^1(c)}{g^1(c)}\ \ \therefore f^1(c)=2g^1(c)$

19. $f(x)=4-(6-x)^{2/3}$ is $[5,7]\Rightarrow\ f(a)=f(5)=4-(6-5)^{2/3}=4-1=3$

$f(b)=f(7)=4-(6-7)^{2/3}=4-(-1)^{2/3}\Rightarrow f(a)\neq f(b)$

Rolle's theorem is not applicable.

similarly LMVT also not applicable.

20. Equation of curve is $\Rightarrow \sqrt{x}+\sqrt{y}=2\Rightarrow \frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{y}}\frac{dy}{dx}=0\Rightarrow \frac{dy}{dx}=\frac{-2\sqrt{y}}{2\sqrt{x}}=\frac{-\sqrt{y}}{\sqrt{x}}$

slope of tangent at (1,1) is $m=-1$

Area of triangle formed by tangent, normal and y-axis =

$\frac{x_1^2(1+m^2)}{2|m|}=\frac{1(1+1)}{2(1)}=1$ sq.units .

21. Equation of curve is $y=be^{-x/a}\Rightarrow \frac{dy}{dx}=be^{-x/a}\left(-\frac{1}{a}\right)=-\frac{b}{a}e^{-x/a}$

for $x=0\Rightarrow y=b\Rightarrow$ slope of tangent $=-\frac{b}{a}$

Equation of tangent is $\Rightarrow y-b=-\frac{b}{a}(x-0)\Rightarrow \frac{y}{b}-1=-\frac{x}{a}\Rightarrow \frac{x}{a}+\frac{y}{b}=1.$

22. $h(x)=\frac{f(x)}{g(x)}$

By applying LMVT

$\Rightarrow \frac{h(p)-h(a)}{b-a}=h^1(c)\Rightarrow \frac{f(b)/g(b)^{-f(a)}/g(a)}{(b-a)g(a)g(b)}=\frac{g(c)f^1(c)-f(c)g^1(c)}{(g(c))^2}$

$\Rightarrow \frac{g(a)f(b)-f(a)g(b)}{g(c)f^1(c)-f(c)g^1(c)}=(b-a)\frac{g(a)g(b)}{(g(c))^2}\Rightarrow k=b-a$

23. Equation of the curve is, $y=ax^2+bx+c$

Points are $(p, ap^2+bp+c), (-p, ap^2-bp+c)$

Slope of the line joining two points $=\frac{2bp}{2p}=b$

$\frac{dy}{dx}=2ax+b; \left[\frac{dy}{dx}\right]_{(\alpha,\beta)}=2a\alpha+b; 2a\alpha+b=b\Rightarrow \alpha=0.$

Level – II

24. A,D	25. A	26. A	27. C	28. B
29.a-q, b-s, c-p, d-r		30. 3	31. C,D	32. a-r; b-p; c-s; d-r
33. 3	34. A, B	35. a-r; b-p; c-s; d-q		

24. Given equations to curves are $y = x^2 + ax + b$ and $y = x(c - x) \Rightarrow y = cx - x^2$

$y^1 = 2x + a, y^1 = c - 2x$

$y^1_{(1,0)} = 2 + a, y^1_{(1,0)} = c - 2$

Curves touch each other at (1,0). Then the slopes of tangents to curves at (1,0) are equal

i.e., $2 + a = c - 2$

(1,0) is point on $y = x^2 + ax + b$ and $y = x(c - x)$

i.e., $0 = 1 + a + b$

$1(c - 1) = 0 = c = 1$

sub (3) in (1),

$2 + a = 1 - 2 \Rightarrow a = -3$

from (2), $a + b + 1 = 0 \Rightarrow -3 + b + 1 = 0 \Rightarrow b = 2.$

25. $m = \left(\frac{dy}{dx}\right)_{(2,3)} = 4$. Equation of normal is $x + 4y - 14 = 0.$

26. $y = x^3 - 2x + 7;$

$y^1 = 3x^2 - 2 \Rightarrow [y^1]_{(1,6)} = 1$

equation of tangent is $y - 6 = 1(x - 1) \Rightarrow x - y + 5 = 0.$

27. $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2;\ (a, b)$

$n\left(\frac{x}{a}\right)^{n-1} \times \frac{1}{a} + n\left(\frac{y}{b}\right)^{n-1} \times \frac{1}{b} y^1 = 0 \Rightarrow \left(\frac{y}{b}\right)^n \cdot \left(\frac{y}{b}\right)^{-1} \times \frac{1}{b} y^1 = -\left(\frac{x}{a}\right)^n \left(\frac{x}{a}\right)^{-1} \times \frac{1}{a}$

$y^1 = -\frac{b}{a} \frac{\left(\frac{x}{a}\right)^n \times \left(\frac{x}{a}\right)^{-1}}{\left(\frac{y}{b}\right)^n \cdot \left(\frac{y}{b}\right)^{-1}} \Rightarrow y^1 \Big|_{(a,b)} = \frac{-\frac{b}{a}(1)^n (1)^{-1}}{(1)^n (1)^{-1}} = -\frac{b}{a}$

Equation of tangent is

$y - b = -\frac{b}{a}(x - a) \Rightarrow ay - ab = -bx + ab$

$\therefore bx + ay = 2ab \Rightarrow \frac{bx + ay}{ab} = 2 \Rightarrow \frac{x}{a} + \frac{y}{b} = 2.$

28. $6y = 7 - x^3;\ (1,1) \Rightarrow 6y^1 = 0 - 3x^2 \Rightarrow y^1 = \frac{-3x^2}{6} = y^1|_{(1,1)} = -\frac{1}{2}$

Equation of tangent is $y - 1 = -\frac{1}{2}(x - 1)$

$2y - 2 = -x + 1 \Rightarrow x + 2y - 3 = 0.$

29. Given point (1,1)

a) $6y = 7 - x^3 \Rightarrow 6y^1 = 0 - 3x^2 \Rightarrow y^1 = \frac{-3x^2}{6} = -\frac{x^2}{2} \Rightarrow y^1\Big|_{(1,1)} = -\frac{1}{2}$

Equation of tangent is $y - 1 = -1/2(x - 1).$

$2y - 2 = -x + 1 \Rightarrow x + 2y - 3 = 0.$

b) $y = 3x^2 + 4x - 6 \Rightarrow y^1 = 6x + 4 \Rightarrow y^1|_{(1,1)} = 10.$

Equation of tangent is $y - 1 = 10(x - 1) \Rightarrow y - 1 = 10x - 10 \Rightarrow 10x - y - 9 = 0.$

C) $x^2 + y^2 = 2 \Rightarrow 2x + 2yy^1 = 0 \Rightarrow yy^1 = -x \Rightarrow y^1 = -x/y \Rightarrow y^1|_{(1,1)} = -1$

Equation of tangent is $y - 1 = -1(x - 1) \Rightarrow x + y - 2 = 0$.

d) $x^2y = 1 \Rightarrow x^2y^1 + y(2x) = 0$

$x^2y^1 = -2yx \Rightarrow y^1 = \dfrac{-2y}{x} \Rightarrow y^1\Big|_{(1,1)} = -2$

Equation tangent is $y - 1 = -2(x - 1) \Rightarrow 2x + y - 3 = 0$.

30. Given $y = x^3$;

$y^1 = 3x^2 \Rightarrow y^1|_{(1,1)} = 3$

$LSN = |y_1 \cdot m| = |1.3| = 3$.

31. $x = 4t^2 + 3, y = 8t^3 - 1 \Rightarrow \dfrac{dy}{dx} = \dfrac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \dfrac{24t^2}{8t} = 3t$

Equation of tangent at the point $(4t^2 + 3, 8t^3 - 1)$ is $y - 8t^3 + 1 = 3t(x - 4t^2 - 3)$

The tangent meets the curve again at $P(4p^2 + 3, 8p^3 - 1)$

$\Rightarrow 8p^3 - 1 - 8t^3 + 1 = 3t(4p^2 + 3 - 4t^2 - 3)$

$8p^3 - 8t^3 = 3t.(4p^2 - 4t^2) \Rightarrow 8p^3 - 8t^3 = 12t(p^2 - t^2)$

$8(p^3 - t^3) = 12t(p^2 - t^2) \Rightarrow 2(p - t)(p^2 + pt + t^2) = 3t(p - t)(p + t)[p \neq t]$

$2(p^2 + pt + t^2) = 3t(p + t) \Rightarrow 2p^2 + 2pt + 2t^2 = 3pt + 3t^2$

$2p^2 - pt - t^2 = 0 \Rightarrow (p - t)(2p + t) = 0 \Rightarrow 2p + t = 0 \Rightarrow p = \dfrac{-t}{2}$.

a. $y^2 = x; x^2 = 6y; (2,2)$

$2yy^1 = 1\ 2x = y^1$

$y^1 = \dfrac{1}{2} y\ y^1 = 2x$

$y^1|_{(1,1)} = \dfrac{1}{2}\ y^1\Big|_{(1,1)} = 2$

$\tan\theta = \left|\dfrac{\frac{1}{2} - 2}{1 + 1}\right| = \left|-\dfrac{3}{4}\right| \Rightarrow \tan\theta = \dfrac{3}{4} \Rightarrow \theta = \tan^{-1}\left(\dfrac{3}{4}\right)$.

2. $x^2y = 1, y = e^{2(1-x)}$;

$x^2y^1 + y(2x) = 0\ y^1 = e^{2(1-x)} \times -2$

$x^2y^1 = -2xy\ y^1 - 2e^{2(1-x)}$

$y^1 = \dfrac{-2y}{x}\ y^1\Big|_{(1,1)} = -2e^{2(1-1)}$

$y^1|_{(1,1)} = -2\ \ y^1|_{(1,1)} = -2$

Slopes are equal.

∴ Curves touch each other.

∴ Angle between curves is zero.

3. $xy = 4, x^2 - y^2 = 15, (-4, -1)$

$xy^1 + y(1) = 0 \qquad 2x - 2yy^1 = 0$

$xy^1 = -y \qquad 2yy^1 = 2x$

$y^1 = -\frac{y}{x}$

$y^1 = \frac{x}{y}$

$y^1|_{(-4,1)} = \frac{-(-1)}{-4} = \frac{-1}{4}$ $y^1\Big|_{(-4,1)} = \frac{-4}{-1} = 4$

∴ Product of slopes $= -1$

∴ Curves touch each other orthogonally.

4. $x^2 = 4y, y^2 = 4x$

$2x = 4y^1$

$2yy^1 = 4$

$y^1 = \frac{x}{2}$

$y^1 = \frac{2}{y}$

$y^1|_{(4,4)} = \frac{4}{2} = 2$ $y^1|_{(4,4)} = \frac{2}{4} = \frac{1}{2}$

$\tan\theta = \left|\frac{2-\frac{1}{2}}{1+1}\right| = \left|\frac{3}{4}\right| \Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right)$.

33. Given $f(x) = \sqrt{x-2}$;

[2,6]

$f^1(x) = \frac{1}{2\sqrt{x-2}}$

By LMVT,

$f^1(c) = \frac{f(6)-f(2)}{4} \Rightarrow \frac{1}{2\sqrt{c-2}} = \frac{2-0}{4}$

$\frac{1}{2\sqrt{c-2}} = \frac{1}{2} \Rightarrow \sqrt{c-2} = 1 \Rightarrow c-2 = 1 \Rightarrow c = 3.$

34. $x^3 - y^2 = 0$

$2yy^1 = 3x^2 \Rightarrow y^1 = \frac{3x^2}{2y}$

$y^1|_{(4m^2,8m^3)} = 3m$

Equation of tangent at p is $y - 8m^3 = 3m(x - 4m^2)$

$y = 3mx - 4m^3$

it cuts curve again at Q

solving (1) & (2),

We get $x = 4m^2, m^2$

put $x = m^2$ in (2), then

$y = 3m(m^2) - 4m^3 = -m^3$

$\therefore Q \equiv (m^2, -m^3)$

Slope of tangent at Q is $\frac{dy}{dx}$

$\left.\frac{dy}{dx}\right|_{(m^2,-m^3)} = \frac{3(m^4)}{2\times(-m^3)} = \frac{-3}{2}m$

slope of normal at Q $= \frac{2}{3m}$.

Since tangent at P is normal at Q.

$\frac{2}{3m} = 3m \Rightarrow 9m^2 = 2 \Rightarrow m^2 = \frac{2}{9} \Rightarrow m = \pm\frac{\sqrt{2}}{3}$.

35. a) Given equation of curve is $y = x^3 + 1 \Rightarrow \frac{dy}{dx} = 3x^2$

Lenth of tangent $= \left|\frac{y_1\sqrt{1+m^2}}{m}\right| = \left|\frac{2\cdot\sqrt{1+3^2}}{3}\right| = \frac{2\sqrt{10}}{3}$.

b) Given equation of curve is $y = x^2 + 1$

$\frac{dy}{dx} = 2x \Rightarrow m = \left[\frac{dy}{dx}\right]_{(1,2)} = 2(1) = 2$.

Length of normal $= \left|y_1\cdot\sqrt{1+m^2}\right| = \left|2\cdot\sqrt{1+2^2}\right| = 2\sqrt{5}$.

c) Given equation of curve is $y = b\cdot e^{x/a} \Rightarrow \frac{dy}{dx} = \frac{b}{a}e^{x/a}$

$m = \left(\frac{dy}{dx}\right)_{(x_1,y_1)} = \frac{b}{a}e^{x_1/a}$

Length of sub tangent $= \left|\frac{y_1}{m}\right| = \left|\frac{y_1}{\frac{b}{a}e^{x_1}a}\right| = \left|\frac{be^{x_1/a}}{\frac{b}{a}e^{x/a}}\right| = |a|$.

d) Given equation of curve is $y = x^3 \Rightarrow \frac{dy}{dx} = 3x^2$

$m = \left(\frac{dy}{dx}\right)_{(1,1)} = 3(1)^2 = 3$

Length of sub normal $= |y_1 \times m| = |1\times 3| = 3$.

WORKSHEET-03

Level – I

1. A	2. C	3. A	4. C	5. B	6. A	7. D
8. B	9. C					

Level – II

1. B	2. A	3. A	4. C	5. D	6. B	7. 6
8. B	9. A	10. C	11. A	12. D	13. 128	14. C
15. A	16. B	17. A	18. A	19. A	20. C	21. 2
22. C	23. B	24. 2	25. C	26. a-r; b-p; c-s; d-q		27. A
28. 1	29. A,B,C,D	30. C	31 .B	32. D	33. a-q; b-s; c-r; d-p	
34. B,D	35.a-r,s; b-r,s; c-p,q,r; d-s					

1. $f(x) = x(\log x)^2$

   $f^1(x) = x \cdot 2\log x \cdot \frac{1}{x} + (\log x)^2 \cdot 1 = 2\log x + (\log x)^2$

   $f^1(x) = 0$

   i.e., $2\log x + (\log x)^2 = 0$

   $\Rightarrow \log x(2 + \log x) = 0$

   $\Rightarrow \log x = -2$

   $\therefore x = e^{-2}$.

   Stationary value of $f(x) = f(e^{-2}) = e^{-2}(\log_e e^{-2})^2 = e^{-2}(-2)^2 = 4e^{-2}$.

2. $f(x) = \sin^3 x + \cos^3 x$

   $\Rightarrow f^1(x) = 3\sin^2 x\cos x - 3\cos^2 x\sin x = 3\sin x\cos x(\sin x - \cos x)$

   $f^1(x) = 0$

   i.e., $3\sin x\cos x(\sin x - \cos x) = 0$

   $\Rightarrow \sin x - \cos x = 0, \sin x = 0, \cos x = 0$

   $\Rightarrow \sin x = \cos x, x = 0^0, x = \frac{\pi}{2}$

   $\therefore x = \frac{\pi}{4}$

   $f(0) = \sin^3 0 + \cos^3 0 = 0 + 1 = 1$

   $\Rightarrow f\left(\frac{\pi}{4}\right) = \sin^3\frac{\pi}{4} + \cos^3\frac{\pi}{4} = \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$

   $\Rightarrow f\left(\frac{\pi}{2}\right) = \sin^3\frac{\pi}{2} + \cos^3\frac{\pi}{3} = 1 + 0 = 1$

   $\therefore$ greatest value of $f(x) = 1$.

3. $f(x) = x^x$

   $f^1(x) = x^x(1 + \log x)$ for decreasing, $f^1(x) < 0$

   $\Rightarrow x^x(1 + \log x) < 0$

   $\Rightarrow 1 + \log x < 0$

   $\Rightarrow \log x < -1$

   $\Rightarrow x < \frac{1}{e} \Rightarrow x \in \left(0, \frac{1}{e}\right)$.

4. Let x and y be the numbers.

   $x + y = 6 \Rightarrow y = 6 - x \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{x} + \frac{1}{6-x} = \frac{6-x-x}{x(6-x)} = \frac{6}{x(6-x)}$

   Let $f(x) = \frac{6}{x(6-x)} = \frac{6}{6x-x^2} \Rightarrow f^1(x) = 6\left(\frac{-1}{(6x-x^2)^2}\right)(6 - 2x)$

   $f^1(x) = 0 = \frac{-6(6-2x)}{(6x-x^2)^2} = 0 \Rightarrow 6 - 2x = 0 \Rightarrow x = 3$

   Minimum value of $f(x) = f(3) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

5. Let area of triangle be $\Delta = \frac{1}{2}bc\sin A$.

   where b and c are constants

$\Rightarrow \Delta^1 = \frac{1}{2}bc\cos A$

$\Delta^1 = 0$

$\Rightarrow \frac{1}{2}bc\cos A = 0 \Rightarrow \cos A = 0 \Rightarrow A = \frac{\pi}{2}.$

6. $y = f(x) = \frac{\log x}{x}$

$\Rightarrow f^1(x) = \frac{x \cdot \frac{1}{x} - \log x \cdot 1}{x^2} = 0 \Rightarrow 1 - \log x = 0 \Rightarrow \log x = 1 \ \therefore x = e$

Stationary point $= (x, f(x)) = \left(e, \frac{\log e}{e}\right) = \left(e, \frac{1}{e}\right).$

7. $f(x) = 9x + \frac{1}{x} \ \Rightarrow f^1(x) = 9 - \frac{1}{x^2} = 0 \ \because f^1(x) = 0$

$\Rightarrow 9 - \frac{1}{x^2} = 0 \Rightarrow \frac{1}{x^2} = 9$

$\Rightarrow x^2 = \frac{1}{9} \ \therefore x = \pm\frac{1}{3}$

$\Rightarrow f\left(\frac{1}{3}\right) = 9 \times \frac{1}{3} + 3 = 6 \Rightarrow f\left(-\frac{1}{3}\right) = 9\left(-\frac{1}{3}\right) - 3 = -6.$

8. $f(x) = x^3 + ax^2 + bx + c \Rightarrow f^1(x) = 3x^2 + 2ax + b$

for increasing, $f^1(x) > 0$

$\Rightarrow 3x^2 + 2ax + b > 0 \Rightarrow \Delta < 0 \Rightarrow 4a^2 - 12b < 0 \Rightarrow a^2 < 3b.$

9. $y = f(x) = x^3 - 3x + 2$

$\Rightarrow f^1(x) = 3x^2 - 3 = 0$

$\because f^1(x) = 0$

$\Rightarrow x^2 = 1 \Rightarrow x = \pm 1 \Rightarrow f(0) = 2, f(-1) = -1 + 3 + 2 = 4 \Rightarrow f(1) = 1 - 3 + 2 = 0$

$\therefore$ Absolute maximum $= 4.$

10. Equation of ellipse is $\frac{x^2}{64} + \frac{y^2}{49} = 1$

comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ \Rightarrow a = 8, b = 7$

The least value of of tangents to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is intercepted between coordinate axes is $a + b = 8 + 7 = 15.$

11. $f(x) = (\sin^{-1} x)^2 + (\cos^{-1} x)^2 \Rightarrow f^1(x) = 2\sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}} + 2\cos^{-1} x\left(-\frac{1}{\sqrt{1-x^2}}\right) = 0$

$\Rightarrow \frac{2}{\sqrt{1-x^2}}(\sin^{-1} x - \text{cis}^{-1} x) = 0 \Rightarrow \sin^{-1} x = \cos^{-1} x$

$\therefore x = \frac{1}{\sqrt{2}} \ \therefore f(x)$ is stationary at $x = \frac{1}{\sqrt{2}}.$

12. $f(x) = \sin x - \cos x - ax + b \Rightarrow f^1(x) = \cos x + \sin x - a$

for decreasing, $f^1(x) < 0 \Rightarrow \cos x + \sin x - a < 0$

$\Rightarrow a > \sqrt{2}.$ ($\because$ max. value of $\sin x + \cos x = \sqrt{2}$)

13. $f(x) = x^2 + (16 - x)^2 = 2x^2 + 256 - 36x = 4x - 32 = 0$

$\therefore x = 8 \Rightarrow y = 8$

$\therefore$ Minimum value of $x^2 + y^2 = 128$.

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$$f(x) = 2x - \tan^{-1} x - \log\left(x + \sqrt{1+x^2}\right)$$

$$f^1(x) = 2 - \frac{1}{1+x^2} - \frac{1}{x+\sqrt{1+x^2}}\left(1 + \frac{1}{2\sqrt{1+x^2}}2x\right)$$

$$= 2 - \frac{1}{1+x^2} - \frac{1}{x+\sqrt{1+x^2}}\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}} = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}}$$

$$= \frac{2(1+x^2) - 1 - \sqrt{1+x^2}}{1+x^2} = \frac{1+2x^2-\sqrt{1+x^2}}{1+x^2}$$

for increasing $f^1(x) > 0 \Rightarrow (1+2x^2)^2 > 1+x^2$

$\Rightarrow 1 + 4x^4 + 4x^2 > 1 + x^2 \Rightarrow 4x^4 + 3x^2 > 0 \Rightarrow x^2(4x^2+3) > 0 \Rightarrow x > 0.$

15. $f(x) = 8x^2 - x^4 - 4$

$\Rightarrow f^1(x) = 16x - 4x^3 \Rightarrow f^1(x) = 0 \Rightarrow 16x - 4x^3 = 0 \Rightarrow 4x(4 - x^2) = 0$

$\therefore x = 0$ or $x^2 = 4 \Rightarrow x = \pm 2$

$f(0) = -4, f(-2) = 12, f(2) = 12$

$\therefore$ stationary points are $(0,-4), (-2,12), (2,12)$.

16. $f(x) = |x-1|$ is not differentiable at $x = 1$ critical point is at $x = 1$.

17. Let x, y be two numbers.

$x + y = 20 \Rightarrow y = 20 - x$

Let $f(x) = x^2(20-x)^3$

$f^1(x) = x^2 \times 3(20-x)^2(-1) + (20-x)^3 2x = (20-x)^2 x[-3x + 2(20-x)]$

$= (20-x)^2 x(40-5x) \Rightarrow f^1(x) = 0 \Rightarrow x = 8 \Rightarrow y = 12$

18. Equation of curve is

$x^{2/3} + y^{2/3} = a^{2/3}$

Let $p(a\cos^3\theta, a\sin^3\theta)bc =$ point on the curve

Distance from origin $= a^2\cos^6\theta + a^2\sin^6\theta$

Let $f(\theta) = \sqrt{a^2\cos^6\theta + a^2\sin^6\theta}$

$$= a\sqrt{\cos^6\theta + \sin^6\theta} = a\sqrt{1 - 3\sin^2\theta\cos^2\theta} = a\sqrt{1 - \frac{3}{4}\sin^2 2\theta}$$

Max. value of $f(\theta) = a$. ($\because$ min. value of $\sin^2 2\theta = 0$)

19. To have minimum area, the point (3,4) is the mid point of potion of line intercepted between coordinate axes $A = (6,0), B = (0,8)$

Equation of line is $\frac{x}{6} + \frac{y}{8} = 1 = 8x + 6y = 48 \Rightarrow 4x + 3y = 24.$

20. Equation of curve is

$2x^2 + y^2 = 2x \Rightarrow y^2 = 2x - 2x^2.$

Let $p(x, y)$ be a point on the curve

distance from $A(a, 0)$ is

$PA^2 = (x-a)^2 + y^2 = (x-a)^2 + 2x - 2x^2$

Let $f(x) = (x-a)^2 + 2x - 2x^2 \Rightarrow f^1(x) = 2(x-a) + 2 - 4x = 2x - 2a + 2 - 4x$

$2x = 2(1-a)$ $\because f^1(x) = 0$

$x = 1 - a$

tangent distance, $\sqrt{f(1-a)} = \sqrt{(1-a-a)^2 + 2(1-a) - 2(1-a)^2}$

$= \sqrt{1 + a^2 - 4a + 2 - 2a - 2 + 4a - 2a^2} = \sqrt{2a^2 - 2a + 1}$.

21. $f^1(x) = x^2 - 1 = 0$ $\therefore x = 1, -1$

for three distinct roots $f(x_1) \cdot f(x_2) < 0$ where x_1, x_2 are roots of $f^1(x) = 0$

$\left(\frac{1}{3} - 1 - b\right)\left(\frac{-1}{3} + 1 - b\right) < 0$ or $\left(b + \frac{2}{3}\right)\left(b - \frac{2}{3}\right) < 0$. $\therefore b \in \left(-\frac{2}{3}, \frac{2}{3}\right)$

22. $R^2 = \frac{h^2}{4} + r^2 \Rightarrow r^2 = R^2 - \frac{h^2}{4}$ volume of cylinder is $V = \pi r^2 h$

$= \pi\left(R^2 - \frac{h^2}{4}\right)h = \pi R^2 h - \frac{\pi h^3}{4} \Rightarrow V^1 = \pi R^2 - \frac{3\pi h^2}{4} = V^1 = 0 \Rightarrow \frac{3\pi h^2}{4} = \pi R^2$

$\Rightarrow h^2 = \frac{4R^2}{3}$ $\therefore h = \frac{2R}{\sqrt{3}}$

$\therefore$ volume is maximum when $h = \frac{2R}{\sqrt{3}}$.

23. Let x be the side of square base and volume, $V = x^2 h \Rightarrow x^2 h = k \Rightarrow h = \frac{k}{x^2}$

Area of that lining is gives x^2

= Base area + Area of 4 wells $= x^2 + 2h(x + x) = x^2 + 4hx = x^2 + 4x\left(\frac{k}{x^2}\right)$

Let $f(x) = x^2 + 4\frac{k}{x} \Rightarrow f^1(x) = 2x - \frac{4k}{x^2} \Rightarrow f^1(x) = 0$

$\therefore 2x = \frac{4k}{x^2} \Rightarrow 2x = 4 \times \frac{x^2 h}{x^2} \Rightarrow \frac{h}{x} = \frac{2}{4} = 1:2$.

24. $f(x) = \frac{|x-1|}{x^2}$

f is not defined for $x = 0$ and $f^1(x) = 0 \Rightarrow x = 2$

$\therefore f$ is not differentiable for $x = 1$

$\therefore$ Critical points are 1,2 .

25. $f(x) = xe^{-x} \Rightarrow f^1(x) = -xe^{-x} + e^{-x} = e^{-x}(1-x)$

$\Rightarrow f^1(x) = 0 \Rightarrow 1 - x = 0 \Rightarrow x = 1 \Rightarrow f(0) = 0, f(\infty) = 0, f(1) = e^{-1} = \frac{1}{e}$.

26. a) $f(x) = x + \frac{1}{x} \Rightarrow f^1(x) = 1 - \frac{1}{x^2}$

for decreasing, $f^1(x) < 0$.

$\Rightarrow 1 - \frac{1}{x^2} < 0 \Rightarrow x^2 - 1 < 0 \Rightarrow (x+1)(x-1) < 0 \therefore x \in (-1,0) \cup (0,1)$.

b) $f(x) = 1 - x^3 \Rightarrow f^1(x) = -3x^2 \Rightarrow -3x^2 < 0 \Rightarrow x^2 > 0$ $\therefore x \in R$

c) $f(x) = xe^{-x} \Rightarrow f^1(x) = -xe^{-x} + e^{-x} \Rightarrow e^{-x}(1-x)$

for decreasing $f^1(x) < 0 \Rightarrow 1 - x < 0 \Rightarrow x > 1 \therefore x \in (1, \infty)$.

d) $f(x) = x\log x$

$\Rightarrow f^1(x) = x \cdot \frac{1}{x} + \log x \Rightarrow f^1(x) < 0 \Rightarrow 1 + \log x < 0 \Rightarrow \log x < -1$.

$\Rightarrow x < e^{-1} \therefore x \in \left(0, \frac{1}{e}\right)$

27. $f(x) = 2x^3 - 3x^2 - 12x + 8$

$\Rightarrow f^1(x) = 6x^2 - 6x - 12 \Rightarrow f^1(x) = 0 \Rightarrow 6x^2 - 6x - 12 = 0$

$\Rightarrow x^2 - x - 2 = 0 \Rightarrow (x-2)(x+1) = 0 \therefore x = 2, -1$

$f^{11}(x) = 12x - 6 \Rightarrow f^{11}(2) = 24 - 6 = 18 > 0$

$\therefore f(x)$ has minimum value at $x = 2$.

28. Let $f(x) = x + \frac{1}{x} \Rightarrow f^1(x) = 1 - \frac{1}{x^2} \Rightarrow f^1(x) = 0 \Rightarrow x^2 - 1 = 0\ x = \pm 1\ f^{11}(x) = \frac{2}{x^3} \Rightarrow$

$f^{11}(1) = 2 > 0 \Rightarrow f(x)$ has minimum value at $x = 1$.

29. $f(x) = x^3 - 6x^2 + 9x + 15 \Rightarrow f^1(x) = 3x^2 - 12x + 9 \Rightarrow f^1(x) = 0 \Rightarrow x^2 - 4x + 3 = 0$

$\Rightarrow (x-3)(x-1) = 0 \ \therefore x = 1,3$

$f^{11}(x) = 6x - 12 \Rightarrow f^{11}(1) = 6 - 12 < 0 \Rightarrow f^{11}(3) = 18 - 12 > 0 \Rightarrow f^{11}(2) = 12 - 12 = 0.$

$f(x)$ is monotonic.

$f(x)$ increases is $(-\infty, 1) \cup (3, \infty)$.

$f(x)$ decrease is $(1,3)$.

It has point of inflection at $x = 2$.

30. $f(x) = \sin x(1 + \cos x) \Rightarrow f^1(x) = \sin x(-\sin x) + (1 + \cos x)\cos x$

$= -\sin^2 x + \cos x + \cos^2 x = \cos 2x + \cos x.$

$f^1(x) = 0 \Rightarrow 2\cos^2 x - 1 + \cos x = 0$

$\Rightarrow 2\cos^2 x + 2\cos x - \cos x - 1 = 0 \Rightarrow 2\cos x(\cos x + 1) - 1(\cos x + 1) = 0$

$\Rightarrow (\cos x + 1)(2\cos x - 1) = 0 \Rightarrow \cos x = -1$ or $\cos x = 1/2 \Rightarrow x = \pi/3.$

31. $y = x^{-x} \Rightarrow \log y = -x\log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = -\left(x \cdot \frac{1}{x} + \log x\right) \Rightarrow \frac{dy}{dx} = -e^{-x}(1 + \log x) \Rightarrow \frac{dy}{dx} = 0 \Rightarrow 1 +$

$\log x = 0 \Rightarrow \log x = -1 \Rightarrow x = e^{-1} = \frac{1}{e} \Rightarrow y = \left(\frac{1}{e}\right)^{-\frac{1}{e}} = e^{1/e}.$

32. Perimeter of sector $= 2r + r\theta = k \Rightarrow r(2 + \theta) = k \Rightarrow r = \frac{k}{2+\theta}$

Area, $A = \frac{1}{2}r^2\theta \Rightarrow \frac{1}{2}\frac{k^2}{(2+\theta)^2}\theta \Rightarrow \frac{k^2}{2}\frac{\theta}{(2+\theta)^2} \Rightarrow A^1 = \frac{k^2}{2}\left[\frac{(2+\theta)^2 \cdot 1 - 2\theta(2+\theta)}{(2+\theta)4}\right] \Rightarrow \frac{k^2}{2}\left[\frac{4+\theta^2+4\theta-4\theta-2\theta}{(2+\theta)4}\right] =$

$\frac{4-\theta^2}{(2+\theta)^4} \Rightarrow A^1 = 0 \Rightarrow 4 - \theta^2 = 0 \Rightarrow \theta^2 = 4 \Rightarrow \theta = 2.$

34. Equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$

Let $p(a\cos\theta, b\sin\theta)$ be one vertex of rectangle,

Length $= 2a\cos\theta$, Breadth $= 2b\sin\theta$ Area $= 2ab.2\sin\theta\cos\theta = 2ab\sin 2\theta.$

Max. area = 2ab sq.units.

35. a) $g(x) = 2\sin x + \cos 2x \Rightarrow g^1(x) = 2\cos x - 2\sin 2x$

$2\cos x - 4\sin x\cos x \Rightarrow 2\cos x(1 - 2\sin x) > 0$

$\Rightarrow \cos x > 0$ and $1 - 2\sin x > 0$

$\Rightarrow -\frac{\pi}{2} < x < \frac{\pi}{2}$ (or) $1 > 2\sin x \Rightarrow 2\sin x < 1 \Rightarrow \sin x < \frac{1}{2} \Rightarrow x < \frac{\pi}{6}$

(or) $\cos x < 0$ and $1 - 2\sin x < 0 \sin x > 1/2 \Rightarrow x > \pi/6$.

b) $f(x) = \cos^2 x + \sin x \Rightarrow f^1(x) = -2\cos x \sin x + \cos x = \cos x(1 - 2\sin x)$.

$f^1(x) = 0 \Rightarrow \cos x = 0$ or $1 - 2\sin x = 0 \Rightarrow \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}$.

c) $f(x) = \tan^{-1}(\sin x + \cos x) \Rightarrow f^1(x) = \frac{1}{1+(\sin x+\cos x)^2} \times (\cos x - \sin x)$

for increasing, $f(x) > 0 \Rightarrow \cos x - \sin x > 0 \Rightarrow \cos x > \sin x$.

d) $f(x) = x^{99} + \sin x - 1 \Rightarrow f^1(x) = 99x^{98} + \cos x$

$f^1(x) > 0 \Rightarrow \cos x + 99.x^{98} > 0 \ \therefore x \in \left(0, \frac{\pi}{2}\right)$.